

Review

Simulation of melt pool behaviour during additive manufacturing: Underlying physics and progress

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ARTICLE INFO

Keywords:

Additive manufacturing (AM) process
Finite-element (FE) modeling
Melt pool dynamics
Laser powder bed fusion
Simulation

ABSTRACT

Reliable computational models of metal additive manufacturing will assist in optimising part quality, and are likely to play a role in component qualification. A key component of these models will be a detailed simulation of flow and heat transfer in and around the melt pool formed as the powder bed is melted. This paper reviews the burgeoning literature concerning melt pool simulation. The physical theory underlying the current benchmark models is first presented and the main approximations and assumptions discussed. The individual capabilities of the leading simulation groups around the world are listed in detail. Publications by less prominent research groups are also summarised. Finally, the overall status of melt pool simulation and the implications for model development are discussed.

1. Introduction

Metal additive manufacturing (AM) refers to the use of the energy provided by a laser, electron beam or electric arc to melt metal powder or wire, building a component over many passes, typically using computer-aided design (CAD) data. Processes are classified according to the form of the metal and the source of the energy. According to the terminology formulated by ASTM International, the relevant categories are powder-bed fusion and directed energy deposition [1]. Powder bed fusion can be divided into selective laser melting, in which regions of a metal powder bed are melted by a moving laser beam in an inert gas environment, and electron beam melting, in which the laser beam is replaced by an electron beam and operation occurs in a vacuum. In both cases, a roller or blade is used to spread additional powder layers over the bed between laser or electron beam passes. Directed energy deposition uses metal wire or powder, deposited or fed from above the component being built; these approaches are known respectively as wire-fed and blown-powder methods. The metal can be melted by a laser, electron beam or plasma arc; in the latter case, only wire-fed approaches have been widely used.

Simulation and modelling of AM processes are being pursued intensively by both researchers and by commercial software providers. The broad objectives are to accurately predict part properties and performance and to understand the sensitivities to important process parameters. As well as assisting in determining parameters for optimum

part quality, it is proposed that reliable models can play a role in process qualification and part certification. A longer term aim is a simulation-based approach to actively control the AM production process, using feedback from process diagnostics.

A full simulation of an AM process can be broken down into a number of sub-models, covering residual stress, microstructure development, powder-bed raking for powder-bed AM or powder trajectories in blown-powder AM, and the detailed behaviour in and around the melt pool [2]. Ultimately, an important aim is to tie these sub-models into a single coupled 'multi-scale' simulation of the AM process, which together with an understanding of the structure-property relations, predicts part properties for given AM process parameters.

A number of useful reviews of AM modelling and simulation have been published in the past few years. King et al. [3] presented a comprehensive review of modelling at both powder and part scales, emphasising their work at Lawrence Livermore National Laboratory (LLNL), but also considering other contributions in some detail. A review of similar scope from the ESI group [4] is also valuable, but is less comprehensive in its survey of other work. Markl and Körner [5] presented a review that avoids any equations and emphasises powder-level approaches and multi-scale considerations. The team at Northwestern University has discussed the issues and difficulties in developing a multi-scale simulation [6,7], with a focus on the need for accurate process-structure-property relationships. A workshop held at Oak Ridge National Laboratory (ORNL) sought to identify the key simulation

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<https://doi.org/10.1016/j.addma.2019.100909>

Received 13 May 2019; Received in revised form 4 October 2019; Accepted 6 October 2019

Available online 25 October 2019

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challenges [8]. Sames et al. [9] from ORNL provided a detailed description of AM metallurgy and processes, with a large section devoted to simulation methods. Schoinochoritis et al. [10] provided a good coverage of AM simulations up to 2014, including a handy table listing papers with validation data, but algorithm detail was limited to thermal stress. Karayagiz et al. [11] incorporated an extensive review of powder-bed fusion simulation as background before describing their own work. A 2012 review by Fan and Liou [12] is beginning to become dated, given the pace of AM simulation developments, but includes relevant sections on solidification and free surface methods. Gong et al. [13] provided a general description of the electron-beam melting process with particular attention to microstructure and a useful summary of reported modelling of the process. A NASA review [14] contains some less-cited references and an informative summary of microstructure evolution studies.

Our review is concerned primarily with modelling of the melt pool, and has two main aims that distinguish it from previous publications. The first is to provide an overview of the state-of-the-art of relevant melt-pool modelling work in greater depth and detail than is possible in a broader review. The second is to assess the different approaches to incorporating the physical effects that arise in melt pool modelling. This includes a critical analysis of the different approaches used to treat important factors such as absorption of the incident laser or electron beam, metal vaporisation rate and heat transfer between particles. As well as AM, we draw upon findings from the related field of keyhole laser welding, for which the simulation understanding is more mature and free of powder complications. We expect this review will be of significant value to researchers developing or assessing melt pool models.

The review is focused on powder-bed, rather than blown-powder or wire-fed, processes. However, many of the problems that arise in powder-bed modelling are common to the other processes. In any case, the vast majority of the literature to date deals with powder-bed processes.

In Section 2, the relevant physics is described in some detail, including the more important of the approximations employed by the various research groups active in the field. In Section 3, the status of the work published by leading research groups is summarised. Section 4 discusses the current status of melt pool simulation and the implications for model development. Conclusions are presented in Section 5.

Some standard abbreviations employed throughout the review are listed here for convenience:

- Processes: EBM – electron beam melting, SLM – selective laser melting, DED – directed energy deposition, PBF – powder-bed fusion.
- Computational methods: FEM – finite-element method, FVM – finite-volume method, CFD – computational fluid dynamics, DEM – discrete-element method, LBM – Lattice-Boltzmann method, SPH – smoothed particle hydrodynamics.

2. Melt pool simulation theory

Models of the melt pool in powder-bed processes consider the processes occurring in the transient melt pool that forms where a laser or electron beam strikes a metal powder bed, and in the immediate vicinity (usually within about 1 mm) of the melt pool. As the beam passes by, it melts the powder, typically composed of $\sim 20\text{--}50\ \mu\text{m}$ metal particles, and creates a liquid melt pool that subsequently cools and solidifies, all of this occurring on a microsecond time scale.

While different computational methods have been applied in modelling the melt pool, solutions to the same fundamental set of physics equations are required, although the formulations used differ in detail. The formulation set out in the following sections is written from the perspective of computational fluid dynamics (CFD), which is the most widely used method for fluid flow modelling in general and for melt

pool modelling in particular. Most CFD codes use either an FEM [15] or FVM [16] approach. Both discretise the model domain into a mesh of small elements or control volumes, respectively, for which the physics equations can be solved in simpler linearized forms before assembly to recreate the whole domain. Traditionally, FEM has been considered to have better handling of complex geometry and sometimes greater accuracy and FVM to be faster and more intuitive for coding new physics. In modern codes, however, these distinctions are very blurred and there is no clear favourite.

Two other computational methods, LBM [17] and SPH [18], have been applied to melt pool simulation. In LBM, rather than solving the Navier-Stokes equations directly, fictive fluid particles are created at a lattice of locations. Their subsequent collisions and movement are tracked and realistic fluid behaviour emerges. SPH is a mesh-free method in which the material is represented by a large set of fictive Lagrangian particles, each defined by a local kernel function. The standard physics equations are applied and decomposed into interactions between neighbouring ‘particles’. There have been few applications of LBM and SPH to melt pool simulation to date. Commercial software is less mature and there are also difficulties in implementing some important physics such as temperature-dependent surface tension and complex boundary conditions. However, in the longer term, both methods show potential due to their implicit handling of complex inter-phase boundaries and their capacity to model the whole-body motion of solid metal particles.

2.1. Simulation requirements

The main requirements of the melt pool model are to:

- Predict the melt pool geometry and temperature distribution;
- Predict formation of defects such as porosity, spattering and other phenomena directly associated with the processes occurring in and close to the melt pool;
- Supply temperature data required to calculate the initial microstructure development near the melt pool;
- Supply an averaged melt pool temperature or heat source to a global model of the thermal and residual stress in the part;
- Provide mesoscale process understanding to inform the approximations made in the global model.

The simulation consists of a coupled solution of fluid flow and heat transfer that incorporates a number of physical effects (Fig. 1):

- Heating and melting of powder particles by the laser or electron beam;
- Surface tension and wetting effects within the partially melted powder bed and underlying substrate;
- Formation of a melt pool by coalescence of molten material;
- Flow within the melt pool generated by thermal buoyancy forces and the Marangoni effect associated with temperature-dependent surface tension;
- Creation, movement and freezing of porosity;
- Calculation of the liquid free-surface profile and its movement;
- Radiative and convective heat transfer from the metal surfaces;
- Evaporation of liquid, creating both recoil pressure on the liquid surface and another source of heat loss;
- Phase change of metal between liquid and solid with associated latent heat release or absorption.

2.2. Fundamental physical equations

All published CFD simulations of the melt pool region in AM start from the following set of simplifying approximations:

- The liquid pool is in a laminar flow regime;

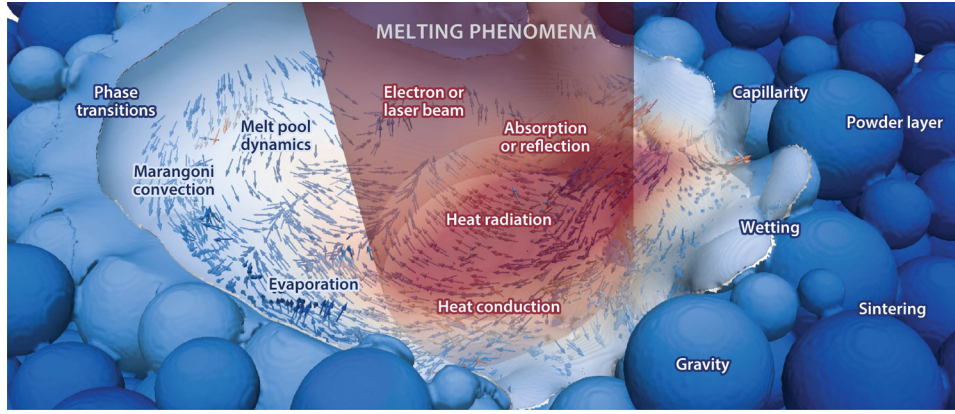


Fig. 1. Main physical processes in melt pool region during powder bed fusion (reproduced from Markl and Körner [5] with permission of ANNUAL REVIEWS; permission conveyed through Copyright Clearance Center, Inc).

- The liquid behaviour is Newtonian and furthermore has no temperature dependence, so the liquid has a constant dynamic viscosity μ ;
- The liquid is incompressible with a constant density ρ . The volume expansion of the liquid is assumed sufficiently small that its only effect is to add a buoyancy force to the momentum equation. The volume thermal expansion coefficient β is usually taken to be constant.

The standard Navier–Stokes equations for mass and momentum conservation [19] then simplify to:

$$\nabla \cdot \mathbf{v} = 0 \quad (1)$$

$$\rho \frac{\partial \mathbf{v}}{\partial t} + \rho(\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla P + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g} - \rho \mathbf{g} \beta (T - T_{\text{ref}}) + \mathbf{S}_m \quad (2)$$

Here P is pressure, $\mathbf{v}$ is the velocity vector and $\mathbf{g}$ is the acceleration due to gravity. The second-last term in eq. (2) is the buoyancy force for which T is the local temperature and T_{ref} is a reference temperature, usually set to the liquid melting temperature. The final term $\mathbf{S}_m$ represents any remaining momentum source terms and will be discussed in section 2.3.

Since the melt pool region has a free surface, an additional equation is needed to track the vapour interface. Most publications use a conventional volume-of-fluid approach in which the fluid volume fraction at a given point, F , obeys the following conservation equation:

$$\frac{\partial F}{\partial t} + \mathbf{v} \cdot \nabla F = 0 \quad (3)$$

The basic equation for energy conservation can be expressed in terms of enthalpy h as:

$$\rho \frac{\partial h}{\partial t} + \rho(\mathbf{v} \cdot \nabla) h = \nabla \cdot (k \nabla T) + S_h \quad (4)$$

Here k is the thermal conductivity and the final term S_h represents remaining enthalpy source terms as discussed in section 2.4.

For the phase change calculation, publications to date assign well-defined liquidus and solidus temperatures T_L and T_S to the metal and assume a linear variation of liquid fraction α between:

$$\alpha(T) = \begin{cases} 0 & \text{if } T < T_S \\ \frac{T - T_S}{T_L - T_S} & \text{if } T_S \leq T \leq T_L \\ 1 & \text{if } T > T_L \end{cases} \quad (5)$$

The temperature dependence of the enthalpy is conventionally expressed as the sum of an integral of specific heat $c(T)$ and a latent heat contribution. In the literature, the latent heat of melting L_m is linearly scaled by the liquid fraction:

$$h(T) = \int_0^T c(T) dT + \alpha(T) L_m \quad (6)$$

The other material properties used in the flow and thermal equations are temperature-dependent but generally straightforward. The exception is the thermal conductivity of the unconsolidated powder, for which a number of models have been employed, as will be reviewed in section 2.5.

The thermal boundary conditions applied by authors to the top surfaces of their models are discussed in section 2.4. On the other model surfaces a variety of conditions have been employed. Most commonly, the sides are made adiabatic, in view of the low conductivity of the loose powder, and the base is set to a fixed temperature. Pseudo-convective or semi-infinite solid approximations have also been used. The temperature fields are initialised to some uniform value, higher in the case of electron beam processes. At best, there is only cursory justification provided with an implicit assumption that the melt pool behaviour is insensitive to the boundary settings chosen.

2.3. Additional momentum source terms

In this section the additional momentum contributions represented by the final term $\mathbf{S}_m$ in eq. (2) are discussed.

2.3.1. Mushy zone flow resistance

In nearly all simulations reviewed, the effect of partial solidification on the metal flow is treated by a mushy zone method. In this approach the mushy zone ($0 < \alpha < 1$) is viewed as a porous medium with liquid and solid phases assumed to be in local equilibrium and exerting a drag force on the flow directly proportional to the local velocity. The constant of proportionality depends on the liquid fraction and the assumed form of permeability. Brent et al. [20] recommended a drag force in the form of the Carman–Kozeny equations and most authors have adopted this approach for AM melt pool simulations. The mushy zone contribution to the momentum source term becomes:

$$\mathbf{S}_m(\text{mushy zone}) = -C[(1 - \alpha)^2/(\alpha^3 + \epsilon)]\mathbf{v} \quad (7)$$

The constant C in eq. (7) is a value set large enough to ensure that velocity drops to zero as the local region freezes, typically 10^5 or greater. The particular value chosen for C can affect the solidification results, but this point has received little attention in the AM literature. An expression relating C to the viscosity and the dendrite cell size d_c of the microstructure,

$$C = 180\mu/d_c^2 \quad (8)$$

is used in modelling of casting and other processes [21]. The constant ϵ is an arbitrary small number incorporated simply to avoid division by zero, ideally as small as numerically possible and typically 10^{-5} or less.

2.3.2. Surface tension

The surface tension in the melt is believed to significantly affect the simulation results in several ways. Surface tension affects the curvature of the liquid interfaces and the shape of pores within the liquid. It leads to balling of the melt pool at large length-to-diameter ratios as described by the Plateau–Rayleigh instability theory. Capillary forces within the partially melted powder bed are thought to be important in the melt movement and coalescence. The Marangoni effect on the melt pool surface (variation of surface tension with temperature and/or species concentration) is one of the main drivers of the liquid flow.

The literature on AM simulation neglects any variation of surface tension with species concentration. The surface tension is approximated to vary linearly with temperature according to:

$$\sigma(T) = \sigma_{\text{ref}} + \frac{\partial \sigma}{\partial T}(T - T_{\text{ref}}) \quad (9)$$

Here, σ_{ref} is the surface tension at some reference temperature T_{ref} (typically the liquidus point) and $\partial \sigma / \partial T$ is the temperature derivative, which is taken to be a constant.

The contribution of surface tension to the momentum equation is expressed in a variety of mathematical forms in the literature according to the particular numerical scheme employed. The most fundamental expression is:

$$S_m(\text{surface tension}) = \sigma \kappa \mathbf{n} + \frac{\partial \sigma}{\partial T} [\nabla T - (\mathbf{n} \cdot \nabla T) \mathbf{n}] \quad (10)$$

Here κ is the surface curvature and $\mathbf{n}$ is the (vector) outward normal to the surface. The first term in eq. (10) is the direct effect of surface tension acting normal to the surface and is often regarded as equivalent to an external pressure. The second term represents the Marangoni effect. Note that the portion inside the square brackets is the component of the temperature gradient parallel to the surface, so the entire term is effectively the spatial gradient of surface tension parallel to the surface.

Another feature of surface tension is the existence of an equilibrium contact angle between liquid and solid. This aspect has received scant attention in the literature but would seem to be quite important. For example, the area of contact between the initial melt and the substrate should affect the rate of cooling. Also, the contact angle between the edge of the melt pool and surrounding frozen material may affect the liquid surface profile, and influences the propensity to balling of the melt pool.

In the dynamic situation of a contact angle θ that differs from the equilibrium value θ_{eq} , a force is exerted on the fluid parallel to the wetted surface, as discussed in detail by Attar [22]. Another source term is added to the momentum equation:

$$S_m(\text{wetting}) = \sigma (\cos \theta - \cos \theta_{\text{eq}}) \quad (11)$$

The practical implementation of surface tension is complicated by the numerical difficulties in accurately defining normals and curvatures of surfaces, which are themselves approximations, based on a VOF interpolation, for example. Rather than attempting to directly prescribe the surface stresses, an equivalent volume force method is generally employed. Schemes are then required to reduce spuriously large velocities ('parasitic currents') in the vapour phase associated with the large density change across the liquid–vapour interface. Application of the contact angle is also difficult. Details of these aspects are beyond the scope of this review but can be explored in several publications [23–26].

2.3.3. Recoil pressure

A recoil pressure is exerted by the flux of atoms escaping as vapour from the liquid metal surface. Klassen et al. [27,28] gave a good description of the theory for electron-beam melting processes, which is summarised here. Consider an alloy composed of N elements $i = 1$ to N , with respective mole fractions χ^i . Atoms of each element vaporise at different rates, with the individual mass fluxes leaving the surface

designated as J_+^i . Some of these atoms recondense, so the nett mass fluxes J_{net}^i that *permanently* escape the surface are smaller quantities equal to $\phi^i J_+^i$, where the factors $0 < \phi^i < 1$ are known as the evaporation coefficients. In this multi-component system the nett mass flux of each element i is:

$$J_{\text{net}}^i = \phi^i \chi^i \gamma_{\text{act}}^i P_s^i(T) \sqrt{\frac{m_A^i}{2\pi k_B T}} \quad (12)$$

Here T is the temperature of vapour leaving the surface (which in practice is taken to be the liquid surface temperature), k_B is Boltzmann's constant, γ_{act}^i is the element's activity coefficient, m_A^i is the atomic mass and $P_s^i(T)$ is the saturated vapour pressure of the pure component at this temperature. Klassen et al. [28] attempted some calculations of separate Al and Ti evaporation from Ti-6Al-4V, but generally authors have approximated alloy evaporation during AM simulation as a pure material with effective physical properties. With this approximation the nett alloy mass flux simplifies to:

$$J_{\text{net}} = \phi P_s(T) \sqrt{\frac{m_A}{2\pi k_B T}} \quad (13)$$

An alternative approach to the mass flux is based on kinetic theory (e.g. Plesset and Prosperetti [29]) and involves two 'accommodation' coefficients for evaporation from a saturated pressure P_s and for condensation from a far-field vapour pressure P_v . In the formulation used by Klassen et al., this complexity is subsumed in the single coefficient ϕ . The evaporation coefficient ϕ effectively incorporates an allowance for the accommodation coefficient for condensation and the ambient pressure, so is not necessarily a constant, as discussed later.

The saturation pressure at a surface temperature T can be calculated in terms of the known saturation pressure P_{atm} at the boiling temperature T_b . First the latent heat of vaporisation L_v is approximated to vary with temperature as:

$$L_v(T) = L_v(0) \sqrt{1 - \left(\frac{T}{T_c}\right)^2} \quad (14)$$

where T_c is the critical temperature. Then the Clausius–Clapeyron equation is integrated from T_b to T to obtain:

$$P_s(T) = P_{\text{atm}} \exp \left\{ -\frac{L_v(0)m_A}{k_B T_c} \left[\frac{1}{T} \sqrt{(T_c^2 - T^2)} - \frac{1}{T_b} \sqrt{(T_c^2 - T_b^2)} + \arcsin\left(\frac{T}{T_c}\right) - \arcsin\left(\frac{T_b}{T_c}\right) \right] \right\} \quad (15)$$

This corresponds to eq. (5) of ref. [27], except that the sign of both arcsin terms has been reversed to correct an error in that article. The sign error is in fact unimportant, because in the AM literature an implicit assumption is made that T and T_b are both so small compared to T_c that the arcsin terms can be dropped and the square roots replaced by T_c . Then eq. (15) simplifies to:

$$P_s(T) \cong P_{\text{atm}} \exp \left[\frac{L_v(0)m_A(T - T_b)}{k_B T T_b} \right] \quad (16)$$

Some checks of these assumptions based on property data for Ti-6Al-4V indicate the associated over-estimation of saturation pressure is small, about 5% for a surface temperature 100 K greater than T_b for example, and less at lower temperatures.

The recoil pressure P_{rec} depends not just on the saturated pressure $P_s(T)$ but also on the condensation rate. Klassen et al. [27] derive an expression for P_{rec} in terms of the evaporation coefficient and the pressure P_{Kn} on the outer side of the Knudsen vapour layer adjacent to the surface:

$$P_{\text{rec}} = \frac{1}{2} P_s + \frac{1}{2} (1 - \phi) \left(\frac{1}{2} P_{\text{Kn}} + \frac{1}{2} P_s \right) \quad (17)$$

They show that for electron beam melting processes, the

evaporation coefficient ϕ is close to 0.82 and that the recoil pressure approaches a limiting (minimum) value of $0.56 P_s$. Then from eq. (16) we have:

$$P_{rec}(T) \cong 0.56 P_{atm} \exp \left[\frac{L_v(0) m_A (T - T_b)}{k_B T T_b} \right] \quad (18)$$

The recoil pressure factor may be significantly larger than 0.56 in non-vacuum conditions because the evaporation coefficient ϕ depends on the Mach number of the vapour at the Knudsen boundary [28]. On the other hand, Semak and Matsunawa [30] quote a private communication from Anisimov that the recoil coefficient does remain close to the minimal value for 'practical values of the ambient pressure'. Eq. (18) has been used in simulations of keyhole laser welding with factors of 0.54 [31–34], 0.55 [35] and 0.6 [36], and with a factor of 0.54 in simulations of laser powder bed fusion [37–41].

To explain observed variations in laser penetration depth with the ambient pressure P_{amb} , Pang et al. [42] recommend that a minimum value of P_{amb} should be imposed on the recoil pressure obtained from eq. (18). They argue for this on the basis of the increased condensation rate caused by the confinement of vapour by an ambient pressure (which is known to increase the recoil pressure factor from 0.56 toward the upper limit of 1) and show that their modification improves the match of their keyhole welding simulations with experiment. However, the authors do not address the issue that their approach implies a recoil pressure would still be present when melt vaporisation is negligible.

2.4. Additional heat transfer source terms

In this section the additional heat transfer contributions represented by the final term S_h in eq. (4) are discussed, all being applied to the top surface.

2.4.1. Evaporative cooling

Evaporation of material from the liquid surface removes significant heat from the melt pool. In section 2.3.3, the rate of mass loss was discussed. Combining eq. (13) and (16), the nett mass flux from the surface, i.e. the balance of evaporation and condensation, becomes:

$$J_{net} = \phi P_{atm} \sqrt{\frac{m_A}{2\pi k_B T}} \exp \left[\frac{L_v(0) m_A (T - T_b)}{k_B T T_b} \right] \quad (19)$$

Note that the evaporation process for alloys is again approximated here as if from a pure material using effective physical properties.

The rate of enthalpy removal from the liquid by evaporation is equal to the mass flux multiplied by the specific enthalpy at the liquid surface. The enthalpy is implicitly removed with the vapour in Lattice-Boltzmann simulations [27,28]. However, in the finite element and finite volume methods used for most melt pool simulations, mass is not explicitly removed from the liquid domain so for correct temperature calculation only the latent heat of vaporisation is applied. The heat loss contribution to the energy equation is thus:

$$S_h(\text{evaporative cooling}) = \phi L_v(T) P_{atm} \sqrt{\frac{m_A}{2\pi k_B T}} \exp \left[\frac{L_v(0) m_A (T - T_b)}{k_B T T_b} \right] \quad (20)$$

Eq. (20) incorporates an allowance for condensation via the evaporation coefficient ϕ . For a vacuum process like EBM, the appropriate value of ϕ is 0.82 [27]. For laser heating, ϕ depends on the vapour Mach number M at the edge of the Knudsen zone [27] and calculations by Knight [43] show that ϕ varies as approximately $0.82\sqrt{M}$ for subsonic flow. However, where mentioned in laser welding and PBF simulations to date [32,33,37,38] a value of 0.82 has been applied, without explanation.

Most publications that include evaporative cooling have used a formulation similar to that above. However, there have been some variations.

Vasquez et al. [44] calculated vapour mass losses during DED by dividing the evaporative heat loss by L_v . The unusual aspect of their simulation was that, rather than applying evaporative heat loss as a boundary condition, they obtained it from the surface temperature gradient and the energy balance required to maintain a steady state. The details are not fully explained but it is inferred that the melt surface was fixed at T_b with an implicit assumption that evaporation is sufficient to suppress any superheating.

Heeling et al. [40] describe a unique method to incorporate evaporative heat loss. Their SLM simulation code considers both fluid and vapour cells in the domain. At each timestep, the next temperature T^n at a melt surface cell is calculated in the normal manner but *without* any explicit evaporation term, and the new recoil pressure P_{rec}^n is determined using eq. (18). Then the hypothetical temperature T that would produce a saturated vapour pressure equal to P_{rec}^n is found from eq. (16). (The justification for this step is not explained.) The difference $(T - T^n)$ is said to represent the evaporative heat loss during the timestep and energy is conserved by converting a corresponding quantity of fluid to vapour in that cell.

2.4.2. Radiative cooling

Most publications include a standard grey-body radiative cooling term in the energy equation:

$$S_h(\text{radiative cooling}) = \sigma \varepsilon (T^4 - T_a^4) \quad (21)$$

Here σ is the Stefan–Boltzmann constant, ε is the surface emissivity and T_a is an ambient temperature, typically set to room temperature. This expression appears to be used throughout the domain with single emissivity and ambient temperature values, even when individual powder particles are simulated. The expression doesn't take into account interception of the emitted radiation by particles in the line of sight, and would therefore overestimate radiative cooling in the powder bed, and around porosity.

2.4.3. Convective cooling

A convective cooling term is usually included in the energy equation:

$$S_h(\text{convective cooling}) = h_c (T - T_a) \quad (22)$$

where h_c is a convective heat transfer coefficient and T_a is an ambient temperature, typically set to room temperature. A range of values for h_c have appeared in the literature; $10 \text{ W m}^{-2} \text{ K}^{-1}$ [12,41,45,46], 15 [47], 20 [48], 25 [49], 40 [50], 50 [51,52], 80 [37,53,54] and 100 [55,56]. There is little or no explanation, except that most of the larger values were intended to compensate for the omission of radiation [54–56]. In any case, convective cooling has much less influence than radiative cooling on the melt pool, and of course none for EBM processes.

2.4.4. Beam heating

The absorption of heat from the laser or electron beam is central to the AM process. Given this, it is interesting and perhaps concerning that many different approaches have been published. To describe the heating methods, it is helpful to first divide the models into two categories; those that model the powder bed as a continuum and those that consider individual particles.

Powder modelled as a continuum

For continuum models, all authors have applied a spatially-varying heat flux with a Gaussian beam cross-section. From that common starting point there are many variants. To describe them more easily, it is helpful to define a generic equation that encompasses the reported heat flux distributions:

$$q(r, z) = \frac{g\eta Q}{\pi R^2(z)} \exp \left[-\frac{gr^2}{R^2(z)} \right] f(z) \quad (23)$$

Here $q(r, z)$ represents a heat flux at radial distance r from the beam axis

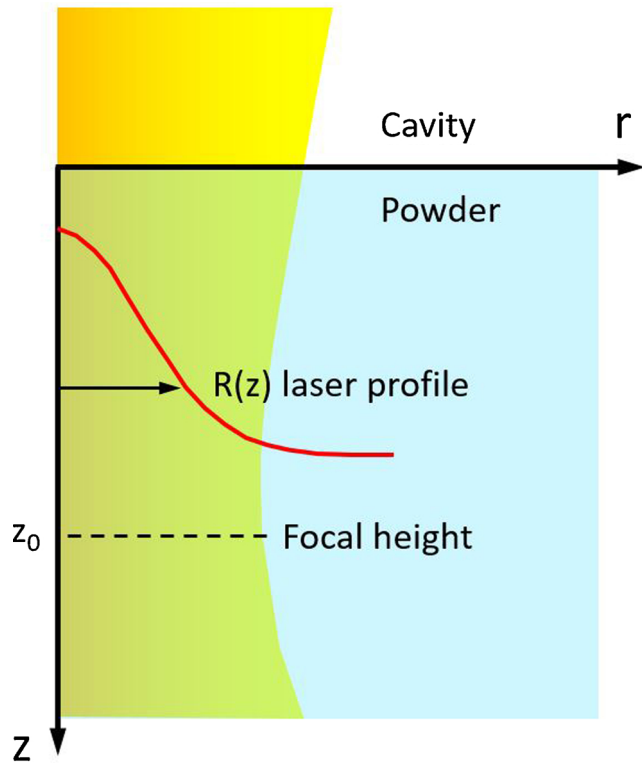


Fig. 2. Schematic of tapered laser beam with Gaussian profile (half-section).

and at depth z from the top of the powder bed. Fig. 2 illustrates the geometry.

The quantity Q is a measure of beam power, such that the area integral of eq. (23) over r should equal the total beam power. The meanings of the other parameters are as follows:

- The quantity $R(z)$ is a representative radius that defines the beam's Gaussian intensity profile at depth z . In virtually all studies, R is treated as a constant, i.e. a collimated beam is assumed. Just two groups [31,57] have modelled a diverging beam by setting $R(z) = R_0 + a|z_0 - z|$, where z_0 is the beam focus location and a is the beam divergence.
- The quantity g is an associated geometric factor that completes the Gaussian profile definition. For laser heating, $g = 2$ has normally been employed as expected for a Gaussian [37,41,45,48,50,54,57–68]. However, Karayagiz et al. [11] used $g = 0.5$, two groups [31,33,34] used $g = 3$, and one [69] omitted squaring the first R and had non-matching g values. One could argue that the factor g depends on the definition of R , but many of the papers do not indicate that this match has been made. For electron beam heating, the radial profile and hence g depend on the filament and deflection coil conditions [70]; two different groups [71,72] have both used $g = 4\ln(10)$.
- The quantity η refers to the material's absorptivity coefficient, i.e. the fraction of the incident radiation that is absorbed rather than reflected. It should be considered in tandem with the function $f(z)$, which defines how the power declines with distance travelled into the bed. Through ray tracing studies, Boley et al. [73] showed how multiple reflections increase the overall absorption of a powder bed compared to a flat surface, by a factor of 1.7 in the case of titanium and larger factors for more reflective materials.

The absorptivity of a given powder material is known to vary with a number of factors such as the nature of the particle surfaces, level of oxidation and laser wavelength. More significantly in terms of

simulation methods, the absorptivity also varies with temperature; in particular the absorptivity decreases markedly as the surface starts to melt [62]. No allowance for this behaviour appears to have been made by any group, both for continuum and discrete powder models. The absorptivity is always regarded as being independent of temperature.

There is a variety of implementations of $f(z)$ in the literature. Most common is to omit any depth dependence $f(z)$ and apply the entire heat flux to the uppermost elements [11,41,44,48,54,58–62,65], with an implied absorptivity coefficient. Other authors prescribe a depth profile but there is little commonality in formulation:

- Mukherjee et al. [74] applied the general Beer–Lambert form of decay through the powder bed, i.e. $f(z) = \exp(-\beta z)$ where β is an optical extinction coefficient. (Strictly, though, the Beer–Lambert law applies to solid material, not a powder.) Additionally, they applied a heat flux to the top of the substrate that was determined by a separate calculation. Yadroitsava et al. [46] also employed an exponential decay.
- Two groups [67,68,75] effectively used $f(z) = 1 - z/s$, where s is a 'penetration depth' at which beam intensity has dropped to less than 1%. This form seems to stem from some EBM measurements [76] although Ladani et al. [68,75] were actually modelling laser heating.
- For EBM, Zah and Lutzmann [72] use $f(z) = 1 + 2z/s - 3z^2/s^2$, which has the problem that it increases initially with depth z . Their equation is sourced from an ARCAM web site which in turn relies on doctoral experimental work by Heinrich in 1973 (see ref. 16 in [72]). Chou and co-workers [45,63,64] also adopted this equation. Jamshidinia et al. [71] referenced Zah and Lutzmann but employed $f(z) = 1 - 2z/s - 3z^2/s^2$, which is problematic since it falls below zero for z much less than s .
- Gusarov et al. [77] applied depth and radial dependences derived from a complex two-flux laser radiation model. The equations can be a good approximation to a full ray-tracing calculation (refer Appendix A) but are lengthy and are so not reproduced here. Their method was adopted by two other groups [40,66].

2.4.5. Powder modelled as discrete particles

There has been a clear shift in recent years from treating the powder bed as a continuum to resolving individual particles. The current section examines how beam heating has been applied to individual particles in melt pool simulations. There have also been many related studies which have focused on determining only the absorption profile through analytic or ray-based methods, without looking at the subsequent melt pool formation. A brief summary of some of those publications is given in Appendix A.

The group headed by Körner at the University of Erlangen-Nuremberg has used Lattice-Boltzmann numerical models to study the melt pool, as distinct from the FEM and FVM models considered in the rest of this section.

- Attar [22] used a 2D LBM. She assumed radiation absorption of a Beer–Lambert form. The incident radiation intensity was given a Gaussian profile, with no depth dependence. Physical particles were modelled, composed of sub-particles by the underlying LBM treatment. The formulation is not made entirely clear, but from the absorption parameters given it would appear that effectively all radiation must have been absorbed by each physical particle encountered, i.e. no transmittance through particles was considered. No reflections were considered.
- In later work by the Erlangen group [78], analytic and experimental models of electron absorption and scattering in a powder bed were assimilated to arrive at (highly complex) expressions for the electron 'dose' vs depth. These dose curve calculations were subsequently applied in LBM models of EBM [27,28] as prescribed heating distributions.

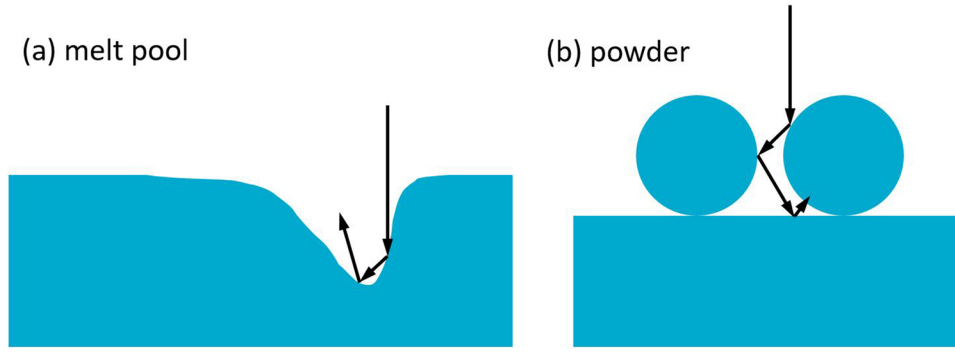


Fig. 3. Examples of enhanced absorption by multiple reflections from (a) melt pool surface and (b) powder particles.

Other authors have used FVM and FEM simulation methods. The heating methods can broadly be sorted in increasing complexity from prescribed profiles, to vertical rays with no reflection, to full ray tracing calculations (Fig. 3):

- Ganeriwala and Zohdi [50] used a combined DEM and finite difference approach; DEM particles for the powder layer and finite difference for the substrate. In the powder bed, each particle was given a lumped heat capacity. Laser power was applied to each particle based on its projected area and a Gaussian power profile as in eq. (23) with an absorptivity η and a depth profile $f(z) = \exp(-\beta z)$. This method is effectively a prescribed heating profile. King et al. [79] also applied an analytic heating profile, but a more sophisticated one, being derived from an analytic radiation model of powder heating [77].
- Lee and Zhang [37] applied a heat flux ‘to the top free surface of the powder layer’ according to a Gaussian power profile and absorptivity η , as per eq. (23) but with no depth function. It is presumed that by ‘top free surface’ they meant the unshadowed particle surfaces since the powder was meshed as particles derived from a DEM calculation. Several other groups [3,38,56,57,80] have used this approach, which can be regarded as ray tracing but without reflections.
- A ray-tracing method has been used for power absorption in several simulations of keyhole-mode laser welding [31,35,36,81], which is not a powder bed process but still relevant. The calculations started with a set of rays whose individual power densities were derived from a Gaussian beam intensity profile. Reflections were treated as specular and the Fresnel equation used to calculate the angle-dependent reflectivity R (and therefore absorptivity $1 - R$ for heating). Each ray was tracked through up to about 15 reflections, until the remaining ray intensity was negligible. Otto et al. [26] also used a ray-tracing method for laser keyhole welding but with two enhancements. One was that the full Fresnel equations were used to separate each ray into parallel and perpendicular polarisations with different reflectivities. The second is a suggestion in their wording that they applied the Gaussian power profile by grading the spatial distribution of the initial rays, which would be more computationally efficient than setting different ray energies.
- Gurtler et al. [69] seem to have been the first to use full ray-tracing for particle heating in AM melt-pool simulation. Like Otto et al., they assumed specular reflections and applied polarisation-dependent Fresnel equations to calculate reflectivities. Within each particle the transmittance was assumed to follow the Beer–Lambert law. Similar ray tracing may have also been applied by Matthews et al. [39], but their description is ambiguous and they may have instead simply applied profile expressions fitted from an earlier detailed ray-tracing study [82].
- Megahed et al. [4] applied a finite-volume formulation of a discrete ordinate radiation model. The radiation transfer equation is

integrated over space by dividing directions into a finite number of discrete solid angles. A key advantage over ray tracing is allowing diffuse reflections. The incorporation into a forerunner of the ESI code has been described by Vaidya [83].

It is important to note that heating a powder bed with reflecting rays is fundamentally different to mimicking those reflections with a prescribed depth profile. A ray-based method is more realistic in that heat deposition is concentrated at the surfaces of particles rather than distributed through their volumes. This means that the surface melting, which greatly enhances heat transfer between particles, is established significantly earlier.

2.5. Effective thermal conductivity of powder

The thermal properties of the powder bed are central to the melt pool development, in particular the thermal conductivity. The background theory and reported methods of implementation are outlined for continuum and discrete models.

2.5.1. Continuum powder models

When modelled as a continuum, the effective thermal conductivity k_{eff} of a loose powder can be considered as the sum of three contributions [75]:

$$k_{\text{eff}} = k_r + k_s + k_g \quad (24)$$

Here k_r represents inter-particle radiation, k_s is the effective conductivity of the solid particle network and k_g represents conduction through the intervening gas (Fig. 4).

The radiative term can be approximated [84] as:

$$k_r = \frac{16}{3} L \sigma_B T^3 \quad (25)$$

Here T is temperature, σ_B is the Stefan-Boltzmann constant and L is the photon mean free path, which to a first approximation is equal to the particle diameter. The radiative term is very small, for example only $0.07 \text{ W m}^{-1} \text{ K}^{-1}$ for $L = 40 \mu\text{m}$ even at a temperature of 1500°C . Nevertheless, it is an important mode of heat transfer for electron beam melting since in a vacuum the k_g term becomes negligible.

In a loose powder, the rate of conduction through the solid material is known to be primarily limited by the size of the contacts between particles. Tolochko et al. [84] performed numerical calculations supported by experiment which gave the following estimate:

$$k_s = \frac{\Lambda a}{r} k_m \quad (26)$$

Here k_m is the metal bulk conductivity, r is the particle radius and a is the radius of connecting necks. The quantity Λ is a geometric factor to account for the degree of coordination of particles, increasing from a minimum of 0.4 for a diamond-like arrangement to a maximum of 1.7 for a BCC arrangement. The problem in practical application of eq. (26)

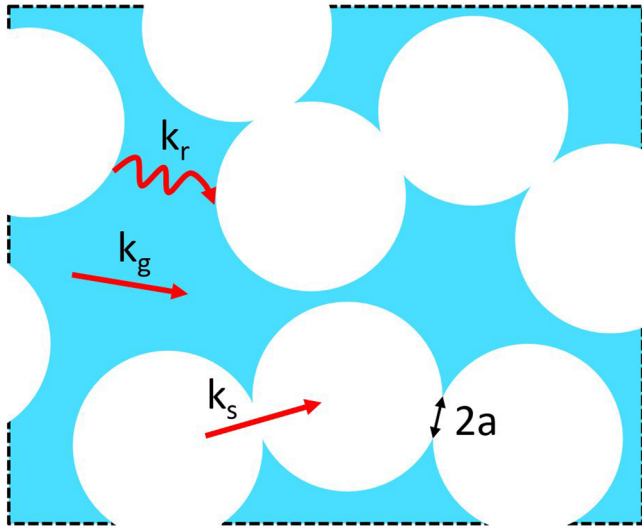


Fig. 4. Portion of a powder bed illustrating three contributions to the effective thermal conductivity.

is the uncertainty in the radius a of the inter-particle contacts. The contact radii are created by gravity-induced deformations [85] and will depend on particle shapes and orientation. For SLM they effectively start near zero but may be more significant for electron beam melting due to sintering produced by pre-heating.

The contribution k_g from gas conduction is also difficult to estimate, and the subject of many theoretical studies, as others have summarised [86–88]. A number of simplifying assumptions are required to even obtain a tractable expression. First it is assumed that the bed consists of spheres, and in a narrow size range. It is also assumed that direct particle–particle conduction is negligible and that the ratio of metal to gas conductivity, k_m/k_{gas} , is larger than about 1000. Note that k_{gas} refers to the gas bulk conductivity, not the *effective* conductivity k_g of gas in a pore network that we are seeking to estimate. The simplest expression for k_g that is applicable to solid volume fractions F_m in the range seen in additive manufacturing is due to Meredith and Tobias [89]:

$$k_g = \frac{(1 + F_m)(2 + F_m)}{(1 - F_m)(2 - F_m)} k_{gas} \quad (27)$$

Note that in eq. (27) k_g includes the effect of conduction through the solid spheres, with a high diffusivity due to the large k_m/k_{gas} limiting value, but omits any sphere–sphere contact. The omission of particle–particle contact conduction justifies applying this expression for k_g in eq. (24).

To give a sense of the magnitudes of the various terms in eq. (27), consider a Ti-6Al-4 V bed at 1500 °C with a volume fraction $F_m = 0.6$. At this temperature, k_m for Ti-6Al-4 V is about 25 W m^{−1} K^{−1} and k_{gas} for argon gas is 0.063 W m^{−1} K^{−1}. Then eq. (27) gives an effective gas conductivity of $k_g = 0.47$ W m^{−1} K^{−1}. (Although the k_m/k_{gas} ratio of about 400 does not strictly meet the $k_m/k_{gas} > 1000$ requirement, it suffices for this rough estimate.)

The gas effective conductivity estimated by eq. (27) is independent of particle size due to the underlying assumptions. More refined treatments consider an additional gas heat transfer mechanism due to free molecular transport in regions where surfaces are closer than the gas mean free path. The mean free path can be calculated theoretically from thermodynamic quantities. Extremely complicated expressions for k_g ensue, such as the Zehner–Schlunder–Bauer (ZSB) model [86,88], which do have a particle-size dependence that may be correlated with observations. A few groups have attempted to incorporate these expressions in SLM simulations, albeit in simplified forms:

- Gu and co-workers at the Nanjing University of Aeronautics and Astronautics [53,59,90–92] simulated laser melting of an AlSi10 Mg powder. Their powder conductivity expression, as explained in earlier work [93], is a variant of the ZSB model for solid spheres surrounded by gas and was developed by Sih and Barlow [94]. In the Nanjing group’s implementation, particle contacts are ignored because terms involving the flattening of contacts between spheres are omitted. Some other groups [54,95,96] have adopted the same method.
- Mukherjee et al. [97] apply an expression developed by Rombouts et al. (eq. 23 in [86]) for the limiting case of zero contact areas, but with an allowance for rapid gas conduction at the contact points. Unlike the Sih and Barlow expression, there is no radiative term.

However, the value of advanced theoretical treatments of powder conductivity is dubious for AM simulations:

- The real powder beds contain particles that are not exactly spherical and in a range of sizes. The inter-particle contact areas are unknown, highly variable but *finite*, especially since there will be some sintering of powder adjacent to a previous beam track, and due to pre-heating in the case of EBM.
- The earlier discussion shows that unsintered powder has very low effective conductivity, at most only a few percent of the bulk value. It is possible that the exact value used in the simulation is unimportant; this will depend on the conditions being modelled and can be investigated by sensitivity studies.
- The effective conductivity increases rapidly where the beam starts to heat and melt the powder. Some type of transition of the effective conductivity to the bulk value needs to be assumed, such as a step change or a ramp over a temperature interval. The form of transition could influence the overall results significantly. However, it is possible that the formation of a melt connection between particles happens so rapidly that through-melt conduction quickly dominates. Then the powder conductivity model can be decided mainly on the basis of numerical robustness.

In contrast to the above theoretical methods, a few authors have attempted to actually measure the effective conductivity of the powder bed to apply in their AM simulations:

- Neira Arce [98] presented data for two types of Ti-6Al-4 V powder at temperatures up to 1600 °C, finding that the values ramped up more steeply above about 700 °C.
- Cheng et al. [63] measured the thermal conductivity of Ti-6Al-4 V powder preserved after EBM preheating and found it increased from 0.63 W m^{−1} K^{−1} at room temperature to 2.44 at 750 °C.
- Andreotta et al. [99] measured the thermal conductivities of Inconel 718 powder beds at three packing densities from 300 K to 1000 K.
- Zhang et al. [100] found that the thermal conductivities of Ti-6Al-4 V and Inconel 625 powders increased from room temperature to 500 °C. However, both powders maintained approximately constant percentages of their bulk values over this temperature range, 4–5 % and 4–7 % respectively.
- Most recently, Wei et al. [101] measured the effective conductivities of five different powders over a range of gas pressures (He, Ar, N₂) and found that the gas contribution k_g dominates over the particle contact term k_s in all cases.
- Moser et al. [85] directly calculated the effective conductivity of a powder using a DEM heat transfer model. Spherical particles were loaded into a virtual box, a heat flux applied and the effective conductivity obtained from the resultant steady-state temperatures. Although subject to several approximations, their results are well-validated and show informative sensitivities to variables such as temperature, depth, gas conductivity, emissivity and packing density. The shortcoming is a lack of predictive ability as the model

would need to be run for each particular bed material.

Given the uncertainties and complications discussed earlier, it is perhaps not surprising that the authors of most melt pool simulations do not even describe their treatment of the powder conductivity. Possibly they simply apply the bulk metal value, raising doubts about the accuracy of their results. A quick summary of those publications which do mention their method is given below.

- For laser melting of Ti-6Al-4 V, Roberts et al. [62] quoted a text [102] suggesting that $k_{\text{eff}} = F_m k_m$. They also reference the work of Rombouts et al. [86] and indicate that k_{eff} should rise from $\sim 0.1 \text{ W m}^{-1} \text{ K}^{-1}$ at room temperature to $\sim 0.3 \text{ W m}^{-1} \text{ K}^{-1}$ around the melting point. They did not state what values they actually implemented, but in another publication [103] mentioned they use a value of $k_{\text{eff}} = 0.25 \text{ W m}^{-1} \text{ K}^{-1}$, presumably a constant. Verhaeghe et al. [66] used a fixed value of $k_{\text{eff}} = 0.20 \text{ W m}^{-1} \text{ K}^{-1}$ for the same process.
- For electron beam melting of Ti-6Al-4 V, Jamshidinia et al. [104] provided a graph and table of their k_{eff} values. They use a fixed value of $0.2 \text{ W m}^{-1} \text{ K}^{-1}$ up to 1000°C and then a ramp up to the melt value of $34.6 \text{ W m}^{-1} \text{ K}^{-1}$. The ramp was applied for numerical convergence reasons, not physical reasons. In fact the full melt value was not applied until several hundred degrees above the liquidus.
- For laser melting of a stainless steel, Gusarov et al. [77] used a fixed small conductivity below the melt temperature, and a step change to the melt value. They cited Rombouts et al. [86] regarding the gas contribution k_g and employed a value of $0.3 \text{ W m}^{-1} \text{ K}^{-1}$ calculated from gas conductivity just below the melt temperature and (presumably) eq. (27) or similar. They stated that the radiative contribution is negligible on the basis of eq. (25).
- Zah and Lutzmann [72] also modelled laser melting of stainless steel. They stated that k_{eff} values calculated from the ZSB model range from 0.03 to $0.1 \text{ W m}^{-1} \text{ K}^{-1}$ below the solidus temperature. However, they did not explain what conductivities were actually applied in their simulation. Both Zhang et al. [47] and Foroozmehr et al. [105] set the powder conductivity to 1% of the bulk value in their simulations of the same process.
- For laser melting of Ti-6Al-4 V, Zeng et al. [58] scaled their powder conductivity as 0.602 times the bulk value. However, this scaling factor is dubious since it was obtained from measurements of already-sintered powder [106].
- For a pure iron powder, Yin et al. [48] set $k_{\text{eff}} = (F_m)^4 k_m$, although in the cited reference [107] the exponent is said to be an empirical parameter of 2–3, not 4. They assumed porosity was constant up to the solidus. Karayagiz et al. [11] applied the same method to a Ti-6Al-4 V powder.
- For electron beam melting of Ti-6Al-4 V, Shen and Chou [67] modelled the powder conductivity as the sum of the k_r and k_s terms in eqs (25) and (26). However, they did not say what neck radius or geometric factor Λ were assumed, or supply any actual powder conductivity values.
- Arisoy et al. [65,108] used $F_m k_m$ when simulating SLM of Inconel 625.
- In their simulation of AM by laser melting, Romano et al. [75] described the powder effective properties in general, including the Eqs (24)–(27) provided earlier. They presented a graph of the powder conductivities they employed for Ti-6Al-4 V, SS316 and Al7075. The equation parameters used to calculate these curves were not provided. Their graph for Ti-6Al-4 V is very coarse but shows a conductivity $< 1 \text{ W m}^{-1} \text{ K}^{-1}$ up to $\sim 1000^\circ\text{C}$ and then ramping linearly to $\sim 5 \text{ W m}^{-1} \text{ K}^{-1}$ by about 1600°C .

Finally, it is noted that the powder density in the simulation must be scaled by the metal volume fraction F_m for correct mass and so that the enthalpy increase of the powder material matches the beam power

absorbed. This is implemented in virtually all published continuum simulations. The difficulty arises when the powder subsequently consolidates to a melt. Ensuring mass, volume and enthalpy conservation across this transition can be problematic, depending on the numerical method employed. In some cases it is achieved by modification of cell fill fractions, or by element deactivations. However, in many publications the issue appears to be left unaddressed, which implies the likelihood of large errors.

2.5.2. Discrete powder models

On the face of it, treating the powder as discrete particles would seem to obviate all the conductivity problems. However, that is not the case. Most authors do not explicitly model the gas as a second phase, and so are relying on point–point particle contacts for heat transfer in the bed. This means the rate of heat transfer may be significantly underestimated. Furthermore, any effects of sintering are missed, and it is more difficult than with a continuum model to apply a workaround such as a spatially varying effective conductivity. Of the groups that have modelled discrete particles, only the following information about powder conductivity considerations has been published:

- Gurtler et al. [69] constructed a powder bed as a regular arrangement of spheres with bimodal sizes. They adjusted the separation of the spheres to control the density, which also affects overlaps and therefore thermal conductivity via contact areas. They stated that ‘Despite the thermal conductivities of our powders at room temperature are overestimated due to geometrical constraints, we point out that their thermal conductivity at much higher process temperatures can safely be assumed to match the experimental values more closely’.
- The LLNL group simulated discrete particles but did not explain exactly how they handle powder conductivity, although some information is provided in a review by King et al. [3]. This mentions a code validation test in which they computed the thermal conductivity of stainless steel powder as spheres in air, and compared the ratio of effective conductivity to air for three packing densities and obtained reasonable agreement with data by Rombouts et al. [86]. It also appears that the LLNL code incorporates the thermal diffusive effect of gas between particles but not gas flow in general: ‘The current powder model ignores the gas dynamics of any trapped background gas, instead modelling only the metal and a “void material”’.
- Description of the powder assumptions employed by researchers associated with ESI is also scarce. From the description in Megahed et al. [4], it seems that a gas phase is included in their simulation. However, it also seems there is no particle–particle contact, so heat transfer must be purely through the intervening gas until the particles start to melt.

3. Summary of published research

This section describes the published research in the field of melt pool simulation for additive manufacturing. In sections 3.1 to 3.4 respectively, research by the four most prominent groups is discussed. These groups have been selected, somewhat subjectively, on the basis of their simulation capabilities, number of authors and publication output. The work of the remaining groups is outlined in section 3.5, with fuller descriptions of some given in Appendix B.

A table of key information is given for each research group. For clarity and brevity, the simulation capability of each group is presented as *gaps* and *enhancements* relative to a fully-capable benchmark. This hypothetical benchmark combines all the features of the current leading groups:

- 3D geometry with powder bed as discrete particles
- Heat transfer including evaporative and radiative losses

Table 1
Summary of publications from ESI group.

Property	Data
Primary institution	ESI Software Germany GmbH
Co-author affiliations	Honeywell Aerospace, University of Swansea, GE, Aachen University, Fraunhofer Institute, Sigma Labs, Stratonics
Publications	[4,109–123]
Representative authors	Megahed, Mindt, N'Dri, Desmaison
Simulation software	ESI_ACE +
Capability gaps, cf benchmark	–
Capability enhancements	Powder spreading by DEM model, incl. multiple layers Powder heating by discrete ordinate radiation model Inclusion of gas phase
Finest resolution (μm)	~ 3 [4]
Processes simulated	SLM, DED
Materials	Ti6Al4V, IN718, stainless steel

- Beam heating with a Gaussian profile and a (feasible) prescribed depth dependence
- Melting (and solidification)
- Calculation of free surface profile of melt pool
- CFD of melt pool flow
- Temperature-dependent surface tension, buoyancy and evaporative recoil

3.1. ESI research group

A number of publications have stemmed from the ESI software company based on a version of their general CFD software called ESI ACE+, which is tailored to AM simulation. They have a leadership position in melt pool simulation from several perspectives: software capabilities, number of industry linkages via consortia, extent of reported validation studies and range of applications.

Table 1 provides a summary. For simplicity the group is described as 'ESI' but in fact most of their publications have been co-authored with the other organisations listed. Compared to the benchmark model described above, their software has no significant deficiencies and several important enhancements, especially the DEM powder spreading capabilities.

A number of the ESI group papers simulate manufacturing problems in detail, such as the effects of powder properties and process parameters on melt pool dimensions [117], porosity [115,116], balling and denudation [111]. Fig. 5 is a typical plot of temperature contours in a powder bed as the laser traverses.

The powder layers are spread by a coating arm according to a full

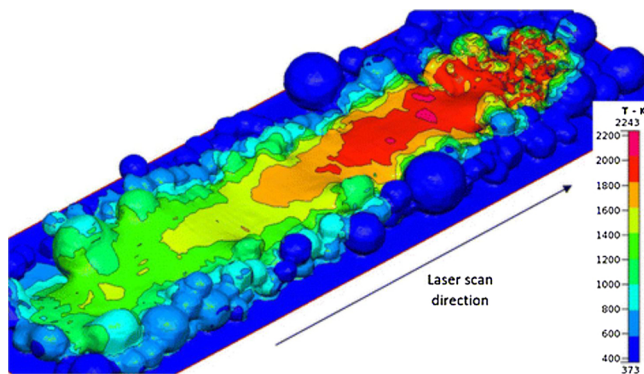


Fig. 5. Example of melt pool simulation from ESI group, showing temperature distribution and surface morphology (reproduced from Megahed et al. [4] with permission <https://creativecommons.org/licenses/by/4.0/>).

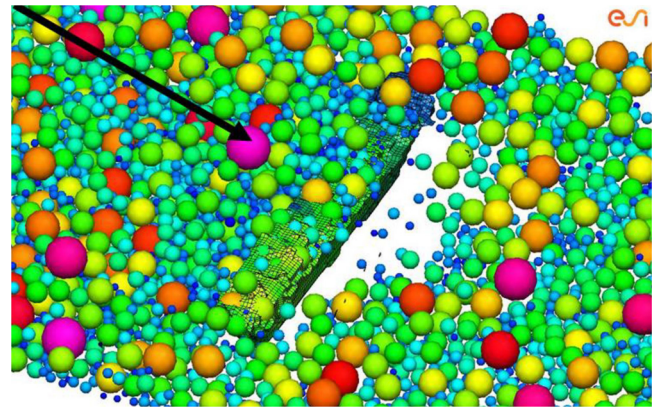


Fig. 6. DEM powder coating simulation, showing spherical steel powder with nine different radii spread over a previously processed layer in the direction of the arrow, with a ridge in the previous layer disrupting the powder flow (reproduced from Mindt et al. [111] with permission of authors).

DEM calculation, including particles distributed across nine bin sizes [119], although still spherical. Impressively, in several studies, re-coating has been performed over the simulation results for previous layers, to look at drag effects and coating arm blockages [111,112,114,119]. Fig. 6 shows a DEM simulation of steel powder in nine spherical sizes spread over a previously processed layer; this example demonstrates how a ridge produces coating problems.

The ESI group has published a large amount of validation data for simulation of distortion of real parts [110–112,122,123].

3.2. LLNL research group

The AM group at the Lawrence Livermore National Laboratory has also been very active in melt pool simulation work. They use in-house software called ALE3D. It has full capabilities relative to the benchmark, except for the fairly minor aspect of not distributing solidification over a mushy zone. A summary is presented in Table 2.

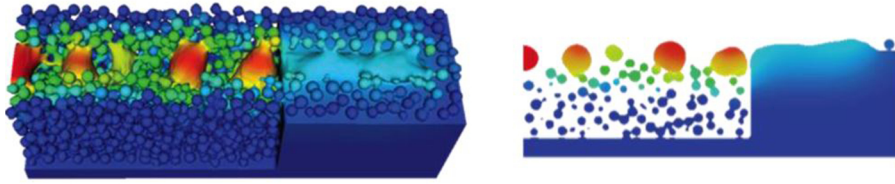
The group has examined the effects of powder and process parameters on the process window and melt pool dimensions [125], porosity [38,130], balling [79], denudation [38,39], overhang effects [79,128] and spatter [38,126]. The latter work on droplet ejection is particularly impressive in terms of both simulation and experiment, shifting the thinking about the ejection mechanism [126] and apparently leading to a patent [124].

The LLNL code does not have the impressive powder spreading capabilities of the ESI group, instead using a rainfall-like forced deposition method with a single size of spherical particles. Their work has also only featured single-track simulations. However, ALE3D is claimed to have superior parallelisation and also features a Lagrangian movement of particles. The description is not completely clear, but this seems

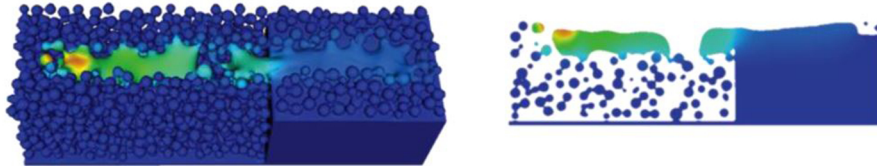
Table 2
Summary of publications from LLNL group.

Property	Data
Primary institution	Lawrence Livermore National Laboratory
Co-author affiliations	Los Alamos NL, Oak Ridge NL, Stanford Synchrotron Radiation Lightsource
Publications	[3,38,39,73,79,124–131]
Representative authors	Khairallah, King, Matthews, Anderson, Rubenchik
Simulation software	ALE3D
Capability gaps, cf benchmark	No mushy zone during solidification
Capability enhancements	Linked Lagrangian particle movement
Finest resolution (μm)	3 [3]
Processes simulated	SLM
Materials	Stainless steel, Ti6Al4V, IN625

Layer 1, speed 2 m/s, 200 W: Severe balling effects



Layer 1, speed 1.6 m/s, 166 W: Discontinuous track



Layer 2, speed 1.6 m/s, 166 W: Continuous track with dross

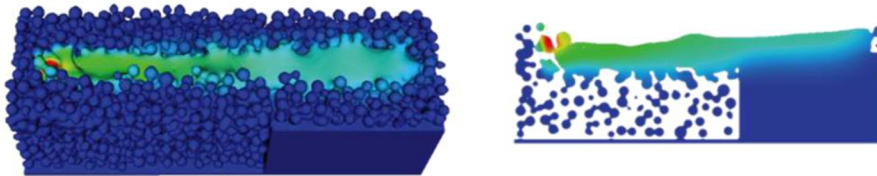


Fig. 7. Example of melt pool simulation from LLNL group (reproduced from King et al. [79] with permission of Informa UK Limited, trading as Taylor & Francis Group, www.tandfonline.com on behalf of Institute of Materials, Minerals and Mining).

Table 3
Summary of publications from the Körner group.

Property	Data
Primary institution	University of Erlangen-Nuremberg (Materials Science and Engineering)
Co-author affiliations	Arcam [133]
Publications	[5,17,22,27,28,78,133–142]
Representative authors	Körner, Klassen, Markl, Attar
Simulation software	In-house
Capability gaps, cf benchmark	2D No solidification shrinkage No Marangoni term No radiation
Capability enhancements	Multiple powder layers Alloy composition spatial distribution
Finest resolution (μm)	1 [28] but usually 5
Processes simulated	EBM
Materials	Ti6Al4V, IN718 [140,141], Ti-48Al-2Cr-2Nb [134]

to involve surface tension forces drawing particles at the melt front into the melt [3].

Their code uses ray tracing for beam heating. Only the initial ray incidence was applied in earlier work but now LLNL researchers apply multiple reflections [128,129,131] using a full ray-tracing code ‘FRED’ as described by Boley et al. [73].

Only SLM studies have been published to date but it does appear LLNL is now also moving to DED as well [127].

An example of the capabilities is shown in Fig. 7, which depicts an investigation of an overhang situation in which the laser travels from right to left over an unsupported region. The top picture shows a melt

balling effect when the laser speed is too high. The central picture shows that the melt layer is more continuous at lower laser speed and the bottom picture shows how a second pass of the laser fills in the gaps.

LLNL is one of the organisations involved in the US DOE Exascale Additive Manufacturing Project, the others being Oak Ridge NL, Los Alamos NL and NIST. It appears that the main simulation codes used in this project all stem from LLNL [132].

3.3. University of Erlangen–Nuremberg research group (Körner)

There are two groups active in AM simulation at the University of Erlangen-Nuremberg. They do not appear to have co-authored any work and use different software; therefore they will be considered separately.

In this section, the work of the group headed by Carolin Körner is discussed. Unlike ESI and LLNL, their work has been in EBM rather than SLM, and their publications indicate a strong theoretical knowledge of the EBM process physics. Unique among AM simulation groups, most of their modelling has used the Lattice-Boltzmann method. The main material considered is Ti-6Al-4 V. The publications are summarised in Table 3.

The group’s model is only 2D, which is a significant limitation. They also omit several important effects (Marangoni, radiation cooling, solidification shrinkage), which is presumed to be due to the difficulty in incorporating them into a LBM simulation. There has been one exception to the 2D limitation; 3D LBM simulations were presented in two linked papers by Ammer et al. [133,136]. However, these were focussed on verification and demonstration, with only a brief experimental comparison [136]. There have been no subsequent 3D LBM studies published. In fact, the Körner group recently instead used a simple FEM

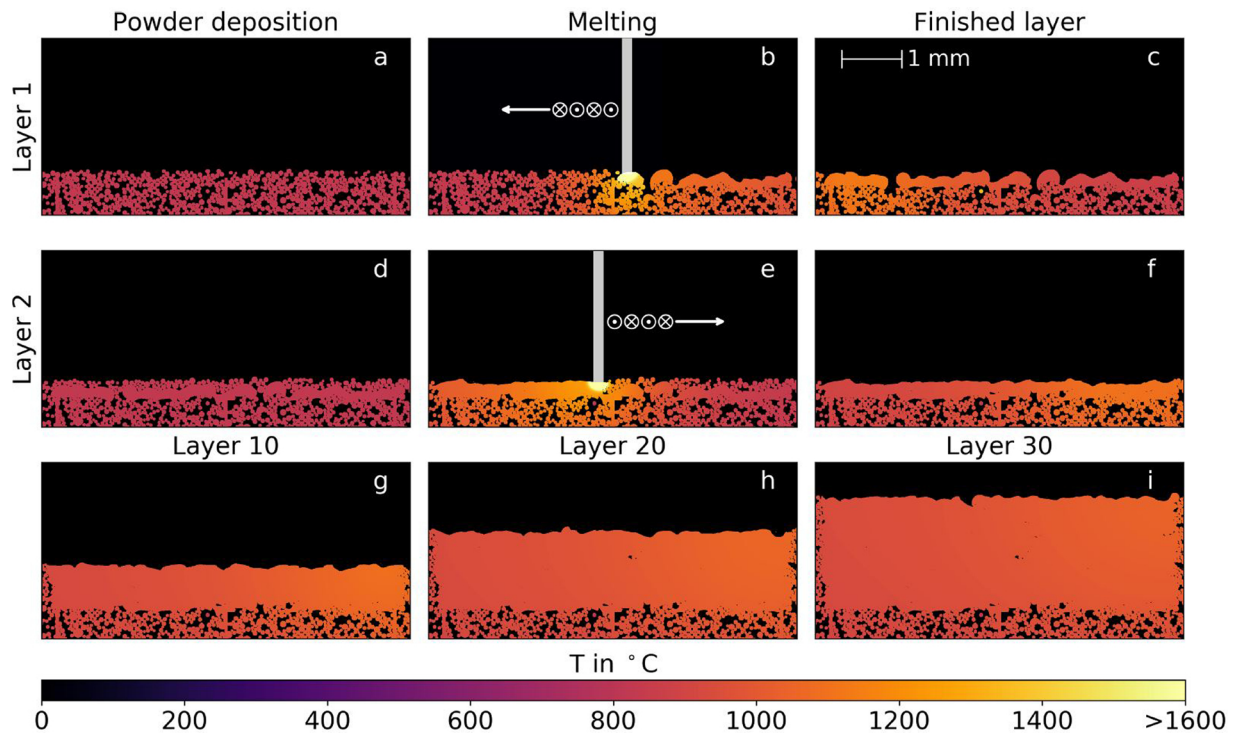


Fig. 8. Example of multilayer simulation from the Körner group (reproduced from Rausch et al. [137] with permission <https://creativecommons.org/licenses/by/4.0/>). Panels (a) and (d) show powder deposited in layer 1 and 2 respectively. Panels (b) and (e) show the melting as the laser is scanned in and out of the page, with translation to the left and right respectively. Panels (c) and (f)-(e) show the finished layer 1, 2, 10, 20 and 30 respectively. Each new layer is coated by ‘rainfall’ deposition. A few ‘pores’ can be discerned after 20 and 30 layers; these types of results are compared by the authors with measurements of experimental cross-sections.

model to study the relationship between melt pool and process parameters and stated this was because a LBM would be too computationally expensive [139].

Much of the group’s research output has concerned verification of their LBM numerical techniques rather than practical applications to AM problems. There were some results related to process optimisation in earlier papers [17,78]. A recent paper [137] by Rausch et al. does have a significant amount of experimental comparison, looking in detail at the effects of beam energy, scan speed and powder bulk density on the number and types of porosity defects.

An impressive aspect of the group’s work has been their use of multilayer powder simulations [78,135,137,142] with up to 100 deposit/melt cycles in the most recent work (as shown, for example, in Fig. 8). Furthermore, they are able to deposit particles in a size distribution (inverse Gaussian). However these capabilities rely heavily on the 2D nature of their simulations, so while the results are certainly useful for qualitative purposes, they may not be fully accurate. As noted elsewhere by the authors [5], the rainfall method employed can lead to non-physical holes in the powder bed. Also, to achieve a designated bulk density they randomly remove particles from the powder bed until the target density is reached [135].

The group has applied its strength in evaporation theory to predict alloy compositional change over repeated melt/cool cycles with good experimental agreement [28] and also extended their model to consider concentration distributions [134], producing plots of Al spatial depletion for Ti-48Al-2Cr-2Nb treated as a binary alloy.

3.4. University of Erlangen–Nuremberg research group (Schmidt)

The second group active in AM simulation at University of Erlangen–Nuremberg is headed by Michael Schmidt. They use a CFD approach rather than the LBM of Körner’s group. Table 4 summarises the key data.

Table 4

Summary of publications from the Schmidt group.

Property	Data
Primary institution	University of Erlangen-Nuremberg (Institute of Photonic Technologies)
Co-author affiliations	University of Vienna, Kazan National Research Technical University
Publications	[26,69,143–146]
Representative authors	Schmidt, Gürtler, Otto
Simulation software	In-house (OpenFOAM)
Capability gaps, cf benchmark	–
Capability enhancements	Full ray tracing, with Fresnel angle dependence
Finest resolution (μm)	Not specified, looks to be about 5μm
Processes simulated	SLM
Materials	Steel, AlSi12Mg

The group has only two publications [69,143] on melt pool simulation for powder-bed melting. In this work the powder bed is built as a regular grid of spheres in two particle sizes. The fractions in each size were varied to match actual powder mixes and the simulation results compared to experimental findings for pool dimensions and final densities. Their simulations have some interesting features:

- To achieve the correct powder bulk density, particles were artificially separated by 2μm. The effective powder bed thermal conductivity was still too large, but further particle separation was rejected as correct density was judged more critical.
- Three adjacent laser tracks were simulated. In some cases, selected particles were removed to gauge the effect of bed ‘defects’.
- Beam heating employed a full ray-tracing calculation (up to 20 reflections) including Fresnel angle-dependent reflectivities.

The powder-bed modelling papers do not specify whether

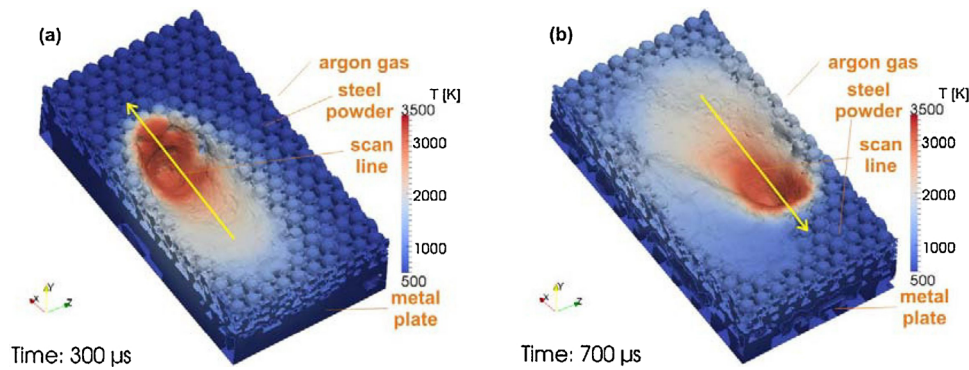


Fig. 9. Example of melt pool simulation from the Schmidt group (reproduced from Gürtler et al. [143] with permission <https://creativecommons.org/licenses/by-nc-nd/3.0/>).

Marangoni and evaporative terms were included, but presumably did because they are described in four earlier papers on laser welding and cutting simulations [26,144–146], which used the same software. These welding studies displayed some impressive capabilities, including tracking of ejected droplets, adaptive meshing and apparently simulation of the vapour plume (the vapour plume appears in the plots but is not mentioned in the text). There is some indication that adaptive meshing may also have been used in the powder-bed work.

In Fig. 9, the regular sphere arrangement used by Schmidt's group is visible. In this example they have simulated a single track and then a second track parallel to the first. As the authors note, it is a highly simplified scenario in which the tracks are quite unrealistic, each being merely a short segment. Nevertheless, a good match with experiment is claimed and results such as these have been used in their two PBF papers to draw a number of conclusions concerning the effects of process parameters.

3.5. Other research groups

The remaining publications in the AM simulation field are briefly described below. Further details are given in Appendix B where noted. As foreshadowed in the Introduction, we focus mainly on the largest segment, namely the simulation of powder bed fusion. These PBF papers are divided into three categories (discrete models, continuum models and continuum models without any CFD) to present them in approximately decreasing order of simulation quality or significance. We conclude by listing the more useful of the DED and laser welding simulations.

The following groups incorporate discrete particles in their powder bed fusion models:

- Ohio State University (Appendix B.1). The main authors Lee and Zhang [37,49,147,148] use the commercial software Flow-3D, which has essentially full melt pool CFD capabilities. The powder bed is created by DEM with a specified PSD (particle size distribution) and spread to 40 μm layer thickness by a cylindrical roller.
- A strong group at Northwestern University led by Gregory Wagner has published CFD models of a variety of AM processes (EBM [70,149,150], SLM [41,57] and DED [151]) based on both in-house and commercial software. Their melt pool simulations include most features with the exception of laser reflections and fully temperature-dependent material properties. They employ a DEM approach to create beds of discrete Ti-6Al-4V particles with a PSD. An EBM study [149,150] included a full spreading calculation which encompassed the effects of both blade angle and spreading over a previously-melted substrate. For SLM [57], particles were introduced by a simpler rainfall method, which was augmented by a coupled air flow calculation to supply particle drag forces.
- Le, Tang and Wong [152] from Nanyang Technological University

also incorporate a spreading step to create a bed of stainless steel particles with a PSD. They compare two different scan patterns in their SLM simulations.

- Wu et al. from Taiwan have published two recent papers on SLM of Ti6Al4V [153] and H13 steel [154] using Flow-3D for a full CFD calculation. The latter paper includes a random powder bed of multiple particle sizes made by the DEM code. The main focus of both papers is to study the surface morphology of the build layers. There is detailed discussion of the effects of evaporative heat loss, although the equation used is not presented.
- Panwisawas et al. [155–158] at the University of Birmingham (Appendix B.2) created a powder bed by a particle-dropping method. They solved the CFD equations using the open-source CFD software OpenFOAM with all physics present but with some simplifications of the evaporative equations. Their work is noteworthy in that the vapour phase flow is said to be included, treated as air, although no further details are provided. A subsequent joint publication between the Birmingham group and Heriot-Watt University [51] investigated powder bed denudation with an axisymmetric model of the vapour plume flow, including multi-species effects, and the powder bed itself approximated as a continuum to act essentially as a vapour source.
- The approach adopted by Ganeriwala and Zhodi [50,159] at the University of California is unique in that the powder bed is created by an impressive DEM calculation but CFD is not included in the subsequent melt pool model. Consequently the particle geometries remain unaltered as they melt, and there is no bed consolidation. Each particle has a uniform temperature and is removed from the calculation domain when it reaches some vaporisation criterion.
- Other published melt pool simulations of discrete particles [56,64,80,160–162] approximate the powder bed as a random or regular array, omit the recoil pressure term and have other limitations (no Marangoni term [56,64,80,160] or 2D [56]).

The following groups treat the powder bed as a continuum but do include CFD of the melt pool:

- A Nanjing University group [53,59,90,91] employed the commercial CFD software Ansys Fluent to simulate SLM of aluminium matrix composites made from AlSi10Mg powders containing AlC or AlN particles. There are some interesting aspects to their work. One paper [91] focuses on surface morphology variations when deliberately operating in a keyhole mode to promote laser absorption, and includes flow of the protective atmosphere above the melt pool in the CFD calculation. Another paper [53] concentrates on the melt pool to simulate the movement of individual AlN particles, and interprets agglomeration effects in terms of final product quality.
- Jamshidinia et al. [71,104] at the Southern Methodist University (Appendix B.3) modelled electron beam melting of Ti-6Al-4V with a

CFD model based on Ansys Fluent. Most physics is included, with the significant omission of free surface movement and evaporative terms. The approximations allowed the model to be run in steady state. The results were linked to a thermal stress calculation.

- A group at Penn State University (Appendix B.4) use what appears to be an in-house CFD model of the melt pool to look at the DED and SLM processes. Their model is less sophisticated than some others but with interesting applications of the results to layer-wise dendrite spacing [163,164] (from cooling rates) and alloy compositional changes [74] (from evaporative losses). Another publication from Penn State University [165] describes attempts to optimise machine control with a process metamodel calibrated by a different (heat-transfer only) melt model. In more recent work they have added a moving grid system for analysis of multiple layers and hatches [97,166].
- Andreotta, Ladani and Brindley from the University of Texas (Appendix B.5) used a model of Inconel 718 powder melting based on the commercial FEM software COMSOL [99]. Their model omitted evaporative effects but did apply an experimentally-determined powder bed thermal conductivity. The primary model is 3D but a separate 2D cross-section model was used to track the cooling and final shape of the melt pool. (The rationale for this approach is not properly explained, and perhaps reflects some software limitation such as speed.) The agreement of melt pool dimensions with experiment is poor. Ladani has also co-authored several publications [68,75,167] describing thermal-only models of powder bed fusion constructed in the commercial FEM software Ansys Mechanical.
- Robichaud et al. [168] incorporate a simplified melt pool CFD model (no recoil, radiation or free surface movement) in their suite of multiscale software and have applied it to Inconel 625.
- An early paper on SLM of Ti-6Al-4V by Verhaeghe et al. [66] omitted the Marangoni effect and recoil pressure but did contain an interesting moving grid method to handle powder compaction.
- For SLM of IN718, Acharya et al. [169] employed melt pool simulation chiefly to provide more accurate temperatures for a phase field calculation of the resultant microstructure.
- In SLM simulations of stainless steel and a nickel-based superalloy IN738LC, Heeling et al. [40] took particular care in their in-house code to correctly handle powder compaction effects when evaluating how laser heat is distributed between powder and melt.
- Although describing laser welding rather than AM, a publication by Ki et al. [170] from the University of Michigan is impressive for a comprehensive coverage of the relevant physics and for combining gas flow above the steel, melt pool flow and multiple laser reflections in a simulation dating back to 2002. They used an in-house finite volume code.

Finally, studies of melt pool formation which only employ a thermal analysis (no CFD) and treat the powder as a continuum are listed briefly below:

- Beuth et al. at Carnegie Mellon University [171,172] used a model of the melt pool constructed using the commercial FEM software Abaqus to find the optimum process window for beam power and speed, for an electron beam wire-fed process. The model was crude but had the distinction of supporting a patent application concerning process window optimisation [173].
- A group led by Brent Stucker from the University of Louisville formed a company selling 3DSIM software for AM simulation, with a number of promotional publications [58,60,61,174,175], since acquired by Ansys.
- Chou's group at the University of Alabama employed Abaqus with user functions for tailored powder conductivity and emissivity [45,63,67,176,177].
- Gusarov et al. [77,178,179] from Ecole Nationale d'Ingénieurs de

Saint-Etienne performed 3D finite difference calculations to support detailed theoretical and experimental analyses of scattering and absorption of radiation by powder beds.

- Zinoviev et al. [52,55] at the University of Bremen also used finite-difference thermal calculations as part of impressive work to simulate grain growth in steel and microstructure evolution in Ti-6Al-4V.
- Zah and Lutzmann from Application Center Augsburg sought to predict melt ball formation and delamination in EBM of steel with a finite-element model [72], the software used being unspecified.
- At the University of Wolverhampton, Roberts et al. [62,103] simulated multilayer thermal histories in Ti-6Al-4V by an element activation approach in the commercial FEM software Ansys CFX.
- Seidel et al. [180,181] from the Technical University of Munich built an Ansys model, which was very coarse as the focus was on multilayer scan path effects.
- A useful evaluation of eight different laser heating approximations was presented by Zhang et al. [47] from the University of Waterloo.
- An early paper by Rouquette et al. [76] from the University of South Brittany featured thermal simulations performed using the Sysweld commercial welding simulation software.
- A number of other papers have used a continuum powder approach with no CFD to look at the sensitivity of SLM to various process parameters [11,46,48,54,65,95,96,105,108,182,183].

DED simulations that incorporate a melt pool solution are less numerous than for PBF. One paper looked at a Ti-6Al-4V wire melted by a plasma arc [184], but the majority consider blown powder deposition. In most cases the deposited powder is steel [120,121,163,164,185–187], but IN718 [148,188], Ti-6Al-4V [12,189,190] and chromium [44] have also been studied. For physics coverage, an early paper by Qi et al. [186] remains close to state of the art as it includes all main liquid terms and also the gas phase, although the particles are treated using analytic expressions. Work by Vogel et al. [120,121] is more comprehensive in terms of the nozzle gas flow analysis and a Lagrangian treatment of the powder stream. Devesse et al. [191] and Kovalev et al. [192] examine the coupling between particle trajectories, heating, laser reflections and attenuation in great detail, but omit melt pool CFD, Kovalev et al. arguing that it is unimportant because the inertia of added particles slows the lateral Marangoni-driven flow.

There is a large body of literature concerning laser welding simulation. The field is longer established than for AM and the lower geometric complexity has enabled more attention to physics details, especially regarding keyhole behaviour. The following papers have been selected as most instructive to the further development of AM simulation capabilities:

- Fottovvati et al. [193] and Fabbro [194] provided excellent reviews of keyhole and melt pool behaviour during deep penetration welding which are also largely applicable to AM.
- High quality simulations have been published including detailed descriptions of the physical equations employed, with particular emphases on evaporation [170], vapour phase dynamics [36], recoil pressure [42] and Mie scattering [34].
- Several papers [31,35,195] explain their advanced implementations of keyhole heating by multiple Fresnel reflections.
- Some authors [23,34] have examined the influence of oxygen absorption on the shape and flow pattern of steel welds via modification of the Marangoni term.
- For welding of titanium, Cho et al. [33] have simulated the melt pool oscillations observed in high speed video images immediately after the laser is stopped.

4. Discussion

Two main trends are apparent in published research on melt pool simulations. The first is the accelerating rate of publication, both in terms of absolute numbers and in the number of active institutions. More than 30 separate bodies published papers in mainstream journals in 2016–2018. Universities and government research organisations still dominate with the few industry partners being limited to aerospace (Honeywell, Pratt & Whitney, Naval Surface Warfare Center). However the proliferation of commercial AM simulation software in the last few years (such as ESI ACE +, Flow-3D Weld, Ansys AM, Autodesk Netfabb, MSC Simufact, Velo3D, Altair Amphyon) may presage more widespread adoption by industry.

The other publication trend is a gradual progression from prototype demonstrations of numerical capability to validation studies and then process sensitivity. Now the most advanced groups are beginning to employ their simulation tools to investigate the physics underlying phenomena such as denudation, spattering and porosity formation. With further maturation the next stages are likely to be practical applications to improve product quality and AM machine design. Initially this may be via simulation-based recommendations but ultimately some form of simulation-derived control embedded into the manufacturing process is anticipated.

The main physics involved in powder bed fusion and subsequent solidification outlined in section 2.1 has been successfully implemented by a number of groups. Nevertheless, there are a number of shortcomings in the work presented even by the leading groups such as ESI, LLNL and University of Erlangen. These can be broadly categorised as model deficiencies and weaknesses in the overall scientific approach.

The ‘model deficiencies’ category refers to the various approximations, errors in key parameters and physical omissions which may reduce simulation accuracy. The main items are summarised below.

- The initial temperature profile from the powder bed to the base of the model is not specified by most authors. When actually described, it seems to be a rough estimate from some heat transfer approximation and not based on actual data.
- The thermal boundary conditions on the sides and base of models are also approximations. Typically the sides are made adiabatic and some simple condition set at the base such as fixed thermal resistance to an ambient temperature.
- A wide variety of beam heating models is employed (section 2.4.4), which reflects the current uncertainties of this critical component. All models omit some important physical effects (temperature dependence of the material absorptivity, the influence of surface melting or oxidation on absorptivity, the relative amounts of diffuse vs specular reflection, interaction of the incoming beam with vapour) and so must be regarded as approximations. Some of these published heating models contain more than 10 adjustable parameters, which is a level of sophistication that is difficult to justify. At the current state of model development, the best that can reasonably be expected is sufficient empirical parametrisation to produce the correct overall heat absorption with a reasonably accurate lateral and depth distribution; a more predictive capability would need to be backed up by considerable experimental validation.
- The codes used to create the initial powder bed range in complexity from simple geometric arrangements, to rainfall models, to full DEM codes, which exhibit the most realistic packing densities. However all published work has used only spherical particles, and generally in a single particle size or a simplified size distribution. Some of the DEM results from ESI are impressive, yet even these are still only beginning to tackle practical effects such as spreading over prior melt and overhangs. This is likely restricted by CPU limitations.
- In terms of material properties, the greatest uncertainty is that associated with the laser or electron beam absorptivity. In simulations employing discrete powder beds (e.g. [3]), it has become clear that

the beam is centred on the melt pool surface rather than on the unmelted particles ahead, except in very low power conditions or at the start of new tracks. However, virtually all simulations to date have used powder rather than melt absorptivities. Absorptivity at high temperatures is a very difficult quantity to measure and reliable data appropriate to AM materials are scarce, especially for the molten state. The powder-bed thermal conductivity also has a large uncertainty. When a continuum approach is used, some theoretical approximations are required (section 2.5.1). When the powder bed is modelled as discrete particles, the effective conductivity is dominated by inter-particle contact areas whose true values are poorly characterised and in any case difficult to numerically enforce (section 2.5.2). More optimistically, though, the most recent data indicate that effective powder conductivities are sufficiently small that melt pool results are probably insensitive to the exact values, except in particular circumstances like overhangs [79].

- There are also uncertainties in vaporisation parameters, especially for SLM where the recoil pressure factor may be underestimated (section 2.3.3) and the evaporative coefficient controlling heat loss may be overestimated (section 2.4.1).
- For the melt pool CFD solution, all authors treat the liquid flow as laminar and incompressible with constant thermal expansion coefficient (for the buoyancy term) and constant viscosity. There has been some justification of the laminar assumption based on low Reynolds number in simulations (for example [104]), although Matthews et al. [39] suggest that vapour emission would create turbulence by a submerged jet mechanism. The other approximations remain largely unexamined. Similarly there has been little investigation of the sensitivity to other important fluid factors such as the wetting contact angle, surface tension, solidification scheme and oxygen contamination. With respect to surface tension, for example, the Marangoni term that drives much of the fluid flow can change sign depending on composition and reverse the flow pattern, as directly visualized in the related field of welding [196].
- Virtually all authors have included radiative heat loss in their models. However the re-absorption of radiation is never mentioned, probably because it is too difficult to properly model. It can therefore be assumed that full heat loss is applied to all free surfaces, including below the powder bed surface and even in enclosed voids. The size of the associated thermal error requires examination.
- Current research involving optical [39,51,126] and X-ray [129,130,197–202] imaging of the melt pool region at high magnification and up to 10 M fps is providing remarkable insights into the fundamental behaviour driving the SLM process. The findings about particle movements, spatter and melt pool dynamics are too complex for a full description here but the following selected highlights give a flavour. First, the powder bed has often been regarded as static but actually considerable movement of particles occurs along the scan path. Metal evaporation creates a plume which directly moves particles and also induces an inward Bernoulli flow of the ambient gas, which draws particles toward the scan path and causes denudation [39]. Schlieren imaging shows the vapour plume angle depends on laser power and scan speed [51]. The gas flows responsible for particle transport are strongly affected by ambient pressure [39,199] and vary with layer number [51]. Spatter is observed to originate from particles entrained by the Bernoulli flow and swept upward by the vapour jet (termed ‘hot’ or ‘cold’ depending whether struck by the laser) plus a smaller amount as droplets ejected from the melt pool by recoil pressure [126,199]. Direct X-ray imaging of the melt pool shows how pores form at the keyhole base [130,197,198,201] and even their subsequent convective migration [202]. X-ray scans also show how laser power and speed influence the keyhole shape and angle [200] and the track surface morphology [202]. Finally, ultrafast X-ray imaging (0.3 μ s exposures at 2 μ s intervals) has revealed the formation and movement of waves across the melt pool surface [129]. None of these

important phenomena are fully described by current models.

The ‘weaknesses in overall scientific approach’ category refers to more general aspects of the published work.

- Very few authors present a complete description of their model setup, omitting a few key items of information regarding boundary conditions, model dimensions or process parameters, meaning that reproduction of their work is not possible.
- Another aspect absent from virtually all publications is any discussion of the mesh sensitivity or convergence criteria of their modelling results. Instead, results for a single mesh size are routinely presented with no indication of numerical error.
- The quality of supporting validation work is variable but generally poor. To some extent that is understandable since it is difficult to obtain in-situ experimental data from commercial AM machines. Validation is therefore mostly of an indirect nature, mainly semi-quantitative comparisons of model results with measurements from the final cooled product. In declining order, the most common types of validation data are melt pool dimensions from single-track experiments, SEM photographs of track surfaces, porosity and microstructure evaluations over part cross-sections and surface roughness. Some groups improve confidence by checking model trends against some process parameter such as laser settings, powder characteristics or number of build layers.

In relation to validation, the publication of additive manufacturing benchmark tests by NIST [203] is an excellent initiative that challenges modellers to test their simulations against well-controlled experimental data. Four groups submitted results incorporating melt pool simulation to the inaugural round of this biennial competition in 2018 [41,168,183,204]. All four selected the same test case, AMB2018-02, which comprised a series of single-scan laser tracks on a bare metal substrate of Inconel 625. Although the simulation results displayed varying levels of success in matching the NIST data for melt pool dimensions, cooling rates and microstructure, all groups must be strongly commended for their participation.

While the long list of model deficiencies might give the impression that melt pool simulation is too inaccurate to be of any use, this is not the case since there have already been many useful findings and insights from published work. There are several reasons for this apparent contradiction.

- It is likely that some of the uncertain parameters, such as some of the thermal boundary conditions and fluid properties, may not actually have a significant effect on key results within their feasible range.
- Although obtaining *ab initio* accuracy for all aspects of a model is not feasible, a pragmatic approach is of value. A model can be developed and applied with specific objectives in mind, and incorporating just the physics necessary to capture the trends of interest. Recognising that all parameters cannot be accurately set, one or two key parameters can be selected and calibrated so that model results match a known validation case within the experimental space of interest. For example, if a particular alloy powder and laser configuration is being investigated, the validation case might be melt pool dimensions for a simple part, with this used to calibrate the laser absorptivity. If this is done carefully, model predictions for the effects of process variables such as laser power, particle shape or particle size can be acceptably accurate.
- In many cases, accurate predictions are not strictly necessary. It is often valuable just to have a firm understanding of trends and be able to make recommendations for process modifications on a scientific rather than *ad hoc* basis.

Although this article is focused on simulation methodology rather

than outcomes, a brief summary of some key results may be useful. The chief value of melt pool simulation studies to date has been in the elucidation and quantification of the physical mechanisms governing AM. The insights gained relate to the fundamental melt pool dynamics, the origin of manufacturing defects and to process optimization. Examples of findings in each of these three broad categories are presented below, with references chosen mainly for their clarity of description.

- Simulation studies have been pivotal to the understanding of the formation and progression of the melt pool during SLM. Heating depends on complex laser reflections through the powder bed [178] at the start of new tracks, but once the melt pool is established the laser is incident primarily on the melt due to the rapid thermal diffusion [3,39]. Evaporative recoil and the Marangoni surface tension term drive the typical surface depression and circulation patterns of the melt [11,40,125]. For EBM, the heating process is perhaps simpler due to reflections being unimportant [70] but the melt pool behaviour is similar [64,134]. For laser cladding, heating depends in detail on the laser scattering and absorption by the powder stream [191,192,205], while the melt flow [148] and bead shape [192] are primarily created by surface tension effects. Simulation work has also led to significant insights into the factors controlling the conduction-keyhole transition [206] (including the type of shielding gas [91]), the depletion of alloy elements by evaporation [134] and associated spatial variations [28], the behaviour of the evaporated plume [51] and the formation of ripples on the solidified surface [57]. As a by-product of model development, dimensionless scaling parameters for SLM and EBM are now available that yield more accurate predictions than the traditional line energy basis [207].
- Melt pool simulations have also been valuable in investigating the origin of various manufacturing defects. The influence of the powder-bed coating has been studied, including effects such as large (spatter) particles creating drag lines and subsequent lack-of-fusion defects [112], the development of local substrate height variations as build layers are added [123] and the variation of build density with the powder PSD [119]. The occurrence of ‘balling’ at low energy densities has been shown to be due to Rayleigh–Plateau instability of the melt pool [72,208] and linked to lack of melting of the substrate [149]. The influence of hatch spacing [166] and scan pattern [150] on voids formed by lack of fusion between layers has been determined. An important mechanism for pore formation due to keyhole instability has been described [38] and a mitigation strategy for such pores at track corners was recently devised [130] based on that understanding. Simulation studies in tandem with experimental work have also helped to identify the causes of spatter from the melt pool [126] and denudation adjacent to tracks [39]. The generation of part surface roughness has been investigated for both EBM [64] and SLM [53,128,150,153,158], including the effects of scan strategy [128,150], beam speed [64], powder layer thickness [158] and the behaviour at unsupported ‘overhangs’ [128].
- Simulation is expected to play an ever-increasing role in process optimization, in the broad sense of improvements to the quality, cost and design flexibility (graded materials, multi-components, etc.) of AM parts. Simulation can assist part quality certification [118,123] and was used to optimise grain orientation in a commercial gas turbine [209]. Melt pool simulation can be used to assist selection of the best production settings through sensitivity analyses [65] and process windows [4,137,171]. However, this is a labour-intensive procedure and the performance measure must be selected from a limited set of output quantities such as bulk density, build rate and track dimensions. Ultimately, we desire an all-encompassing ‘digital twin’ of the AM process [210], operating at multiple scales from microstructure to whole part distortion, with integrated material properties and bidirectional coupling between sub-models [6,7].

Melt pool simulation will be a crucial sub-model of a digital twin, initially in a simplified or sampled manner until computing power increases. The thermal histories supplied by melt pool simulation are essential for microstructural predictions [127] of results like dendrite [169] and grain growth [52,140,168]. Part stress and distortion analyses benefit from the more accurate heat input calculation afforded by melt pool simulation [92,110,211,212].

5. Conclusions

Simulation of the melt pool in metal AM is a relatively young but rapidly growing area of research. A melt pool model is an important component of a full simulation of the AM process, since it allows prediction of the dimensions and properties of the solidified region after each pass of the laser or electron beam, as well as spatially and temporally resolved distributions of temperature in the component being produced. These data are critical in determining properties such as microstructure and residual stress.

The great majority of melt pool models have been developed for powder-bed processes. Several groups have developed models with impressive capabilities, for example prediction of droplet ejection and denudation, multiple laser or electron-beam passes, and porosity formation. From this point of view, progress in melt pool modelling can be seen as impressive. Nevertheless, there are several important problems that are yet to be resolved.

These problems arise because of the complicated physics underlying the development of the melt pool. Some of the phenomena that have to be included in the model include absorption of the laser or electron

beam by the powder, vaporisation of the molten metal and heat transfer between the particles. Full treatments of these phenomena are either not possible or computationally expensive, so approximate treatments are generally used. The sensitivity of the predictions of the model to the choice of treatment is still an open question, and requires careful testing and benchmarking.

Model validation is another area in which additional effort is required, and detailed in-situ time-resolved measurements of temperature and other melt pool parameters would be of great value.

Full predictive simulations of AM processes will require linking melt pool models to other models, including those predicting powder raking or deposition, microstructure formation, and residual stress and distortion. Some steps in this direction have been made, for example through the consideration of several passes of powder coating and melting. However, there are significant challenges to be overcome, including those posed by the large computational resources required by simulations of the melt pool models and the other features of AM processes, the need to couple models across different time and spatial scales, and the complexity of most AM components.

Acknowledgments

The authors are grateful to Drs Vu Nguyen, Paul Cleary, Sharen Cummins, Gary Delaney, Dayalan Gunasegaram and Patrick O'Toole of CSIRO for valuable discussions and comments. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Appendix A. Radiation absorption by powder bed

The transmission and associated absorption of an electron or laser beam through a powder bed is a key physical process required in melt pool simulation. Many theoretical and experimental studies have been devoted to this aspect. For convenience, the main studies referenced by AM simulation publications are briefly summarised here. Studies in which the AM melt pool was considered were discussed in Section 2.4.4.

Rombouts et al. [213] considered light extinction in powder beds, presenting a good general discussion of fundamental mechanisms and theories. Experimental measurements for stainless steel were also presented.

Gusarov and Kruth [214] calculated laser transmission through a powder bed using a complex analytic approach based on the two-flux method, starting from the radiative transfer equation. They presented graphs showing predictions of the deposited energy for both specular and diffuse radiation. The theory assumes opaque particles and so is valid for SLM but not EBM. The equations developed are restricted to a single particle size, although not necessarily spherical, and can be extended to mixed powder materials. The results agreed with full ray-tracing calculations for Fe-Cu and WC-Co mixed powder beds within about 5 %, except near the top surface where the approximation of constant porosity breaks down.

Zhou et al. [215] used a Monte-Carlo ray tracing method to calculate the absorption of radiation by a random bed of bimodal spheres whose distribution was obtained by the rainfall method. Both specular and diffuse cases were modelled, with only a small difference found in the transmittance, but effects of polarisation were not considered. The specular model was validated against measurements performed with polished carbon steel spheres.

Boley et al. [73] from the LLNL group presented a laser ray tracing study using FRED software with user routines. Absorptivity was calculated with the Fresnel formulas. No temperature dependence was considered, but the effect of polarisation was included. Multiple reflections were tracked until 0.1 % of the incident intensity remained. From 50 000 to a few million rays were needed. Tables were provided for different materials for a double layer of regular spheres over a substrate. It was found that most light was absorbed in the top layer. Results were also presented for bimodal random beds, showing the absorption variability along the laser path. An earlier publication from Boley and Rubenchik [82] dealt with laser absorption of fibres imbedded in an epoxy matrix by ray tracing. They also developed a two-flux model and obtained its coefficients by correlation with their ray tracing work. The LLNL group used this approach, of volumetric energy deposition by a two flux method calibrated by prior ray tracing analysis, in their earlier melt pool simulations [39,79,125]. In their more recent work, energy is deposited by direct ray tracing but omitting multiple reflections [3,38].

Moser et al. [216] used Monte Carlo ray tracing to track the absorption in selective laser sintering, with a Gaussian profile of the incident beam. Both specular and diffuse reflections were considered, but polarisation effects were neglected. Randomly packed beds of spheres were used; these were of uniform size in most cases, but there was some consideration of the effect of a size distribution. The package MFIX was used to generate particles by DEM. Extensive results for extinction and absorption coefficients were presented, and it was found that extinction coefficients for specular and diffuse radiation were quite similar.

Han et al. [217] compared efficiency of different ray tracing algorithms in a study of laser keyhole welding. There is no direct application to melt pool simulation, but the paper is a useful resource for ray tracing theory. Laoui et al. [218] presented an analytic ray tracing model, with specular reflection from a random arrangement of spheres considered. The predicted absorption was validated against experiment, as was the width of laser sintered tracks obtained using a heat transfer model.

Klassen et al. [78] considered transmission and absorption of electron beams, and provided a useful review of analytic and empirical expressions. They apply a derived depth profile in their LBM simulation of the powder bed, and claimed good agreement with experiment.

Appendix B. Other simulation research groups

In the main body of the review, the leading research groups in the area of melt pool simulation were described (Sections 3.1–3.4) and the remaining less prominent groups briefly listed (Section 3.5). In this Appendix further details of some of these other groups are provided. The same table format as described in Section 3 is employed.

B.1 Ohio State University

The authors are using Flow-3D with essentially full melt pool CFD capabilities. The powder bed is created by DEM (Table B1).

Table B1
Summary of melt pool simulation publications from Ohio State University

Property	Data
Primary institution	Ohio State University
Co-author affiliations	Los Alamos National Laboratory, Oak Ridge National Laboratory
Publications	[32,37,49,147,148,188]
Representative authors	Lee, Zhang, Farson
Simulation software	Flow-3D
Capability gaps, cf benchmark	Heating only to top surface
Capability enhancements	Powder coating by DEM model, incl. size distribution and multiple layers
Finest resolution (μm)	3 [147]
Processes simulated	SLM, DED
Materials	IN718

B.2 University of Birmingham

This group create a powder bed by dropping particles, with an unusual method involving counting the number of obstacles encountered. They solve the CFD equations using OpenFOAM with all the main physics included, but with significant simplifications of the evaporative equations to ignore back condensation and to set the recoil pressure equal to the vapour pressure. Their work is noteworthy in that the vapour phase flow is said to be included, treated as air, although no modelling details are provided.

A joint publication with the Heriot-Watt University [51] investigated powder bed denudation with an axisymmetric model of the vapour plume flow, including multi-species effects, and the powder bed itself approximated as a continuum to act essentially as a vapour source. This paper includes links to impressive high speed videos of particle movements near the laser beam with an excellent analysis (Table B2).

Table B2
Summary of melt pool simulation publications from University of Birmingham

Property	Data
Primary institution	University of Birmingham
Co-author affiliations	Heriot-Watt University
Publications	[51,155–158]
Representative authors	Panwisawas, Qiu, Ward, Attallah
Simulation software	OpenFOAM
Capability gaps, cf benchmark	Evaporative equation approximations
Capability enhancements	Powder coating by DEM model Vapour included as second phase (but unclear) Linked microstructural calculation
Finest resolution (mm)	5
Processes simulated	SLM
Materials	Ti6Al4V [155,156,158], Steel [51]

B.3 Southern Methodist University

Jamshidinia et al. [71,104] at the Southern Methodist University modelled electron beam melting of Ti-6Al-4V with a CFD model based on Ansys Fluent. Most relevant physics was included, with the significant exceptions of free surface movement and evaporative terms. The approximations enabled running the model as steady state, and the results were linked to a thermal stress calculation (Table B3).

Table B3
Summary of melt pool simulation publications from Southern Methodist University

Property	Data
Primary institution	Southern Methodist University
Co-author affiliations	–
Publications	[71,104]
Representative authors	Jamshidinia, Kong, Kovacevic
Simulation software	Ansys Fluent
Capability gaps, cf benchmark	Powder bed as continuum No evaporative recoil or heat loss No free surface movement Steady state
Capability enhancements	
Finest resolution (mm)	40
Processes simulated	EBM
Materials	Ti6Al4V

B.4 Pennsylvania State University

A group at Penn State University use what is presumed to be an in-house CFD model of the melt pool to look at the DED and SLM processes. Although the physics coverage is less complete than some other groups, that has allowed a moving grid approach to model multiple layers and hatches [97,166] with a view to AM ‘digital twin’ possibilities [210] (Table B4).

Table B4
Summary of melt pool simulation publications from Pennsylvania State University

Property	Data
Primary institution	Pennsylvania State University
Co-author affiliations	Indian Institute of Technology
Publications	[74,97,163–166]
Representative authors	DebRoy, De, Mukherjee
Simulation software	In-house
Capability gaps, cf benchmark	Powder bed as continuum No evaporative recoil or heat loss No free surface movement No radiative heat loss
Capability enhancements	Three layers of deposition, moving grid
Finest resolution (mm)	~40
Processes simulated	DED, SLM
Materials	Ti6Al4V, Steel, IN718, AlSi10Mg

B.5 University of Texas

Andreotta, Ladani and Brindley from the University of Texas used a model of Inconel 718 powder laser melting based on Comsol [99]. Their model omits evaporative effects but does apply an experimentally determined powder-bed thermal conductivity. The primary model is 3D but a separate 2D cross-section model is used to track the cooling and final shape of the melt pool, and includes surface tension and a gas phase. (The rationale for this segregated approach is not properly explained, but probably reflects some software limitation such as speed.) The agreement of melt pool dimensions with experiment is fairly poor.

Ladani also co-authored several earlier publications [68,75,167] describing thermal-only models of powder bed fusion of steel and Ti-6Al-4V constructed in Ansys Mechanical. The table parameters refer to the CFD model (Table B5).

Table B5
Summary of melt pool simulation publications from University of Texas

Property	Data
Primary institution	University of Texas
Co-author affiliations	University of Connecticut, Pratt & Whitney
Publications	[68,75,99,167]
Representative authors	Ladani, Romano, Brindley
Simulation software	Comsol
Capability gaps, cf benchmark	Powder bed as continuum No evaporative recoil or heat loss No convective cooling Free surface movement in a separate 2D model
Capability enhancements	
Finest resolution (mm)	40
Processes simulated	SLM
Materials	IN718

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